
Multimodal Autonomous AI Agents

Russ Salakhutdinov

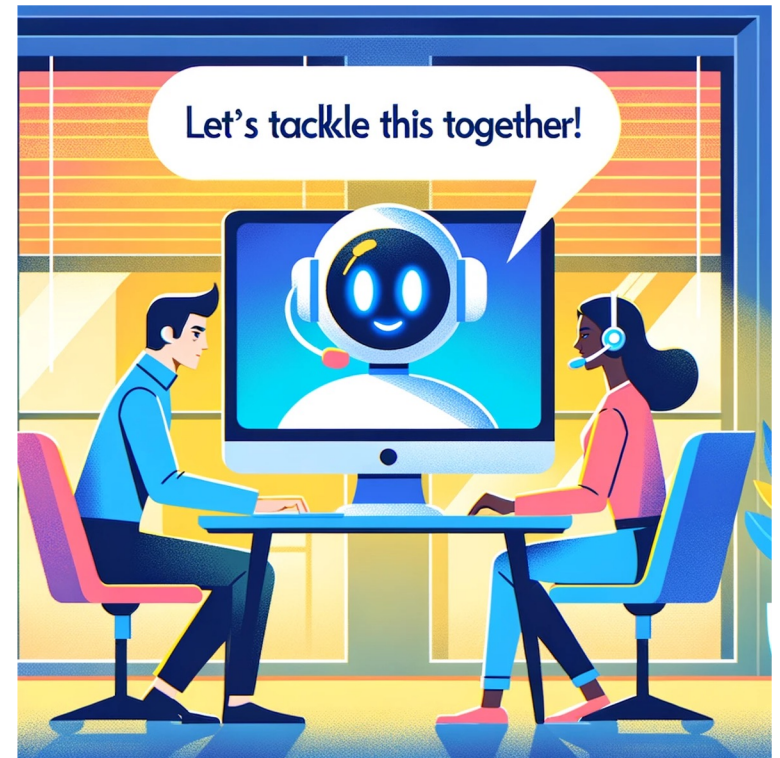
Machine Learning Department
Carnegie Mellon University

**Carnegie
Mellon
University**



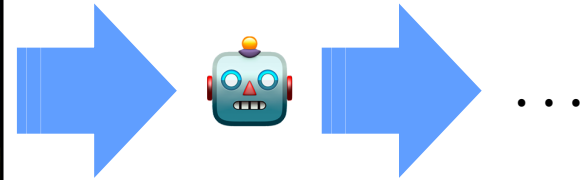
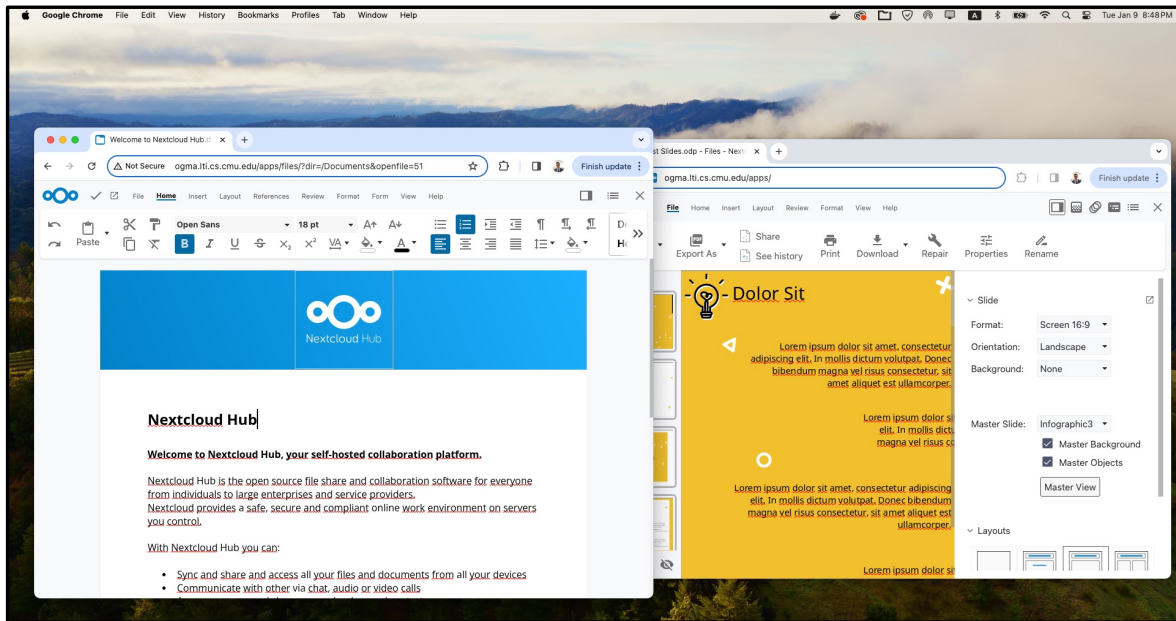
Autonomous AI Agents

- Many productive tasks we perform today are done on the computer
 - And many of these are on the web
- Many opportunities to automate menial tasks
- Augment human capabilities



Generated with DALLE

Autonomous Agents



Task: “Create a set of PowerPoint slides to present the content in this paper.”

Task: Navigate to a page of a good Thai restaurant in Pittsburgh. It should have at least 200 reviews and 4.3 stars. Pick the one with the highest rating

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(venv) jingyuk@GS22433 visualwebarena %  
python run_demo.py \  
  --instruction_path agent/prompts/jsons/  
  /p_som_cot_id_actree_3s.json \  
  --start_url "https://www.yelp.com/" \  
  --intent "Navigate to the page of a go  
od Thai restaurant in Pittsburgh. It sho  
uld have at least 200 reviews and 4.3 st  
ars. Pick the one with the highest ratin  
g." \  
  --result_dir demo_test_yelp \  
  --model gpt-4-vision-preview \  
  --action_set_tag som --observation_ty  
pe image_som \  
  --render
```


Task: Make a reservation at Pusadee's Garden for 2 people on the earliest date for dinner. Use my name JY Koh and phone number 650-555-5555.



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  --start_url "https://www.google.com/" \  
  \  
  --intent "Make a reservation at Pusadee's Garden for 2 people on the earliest  
date at any time. Use my name JY Koh and  
phone number 650-555-5555." \  
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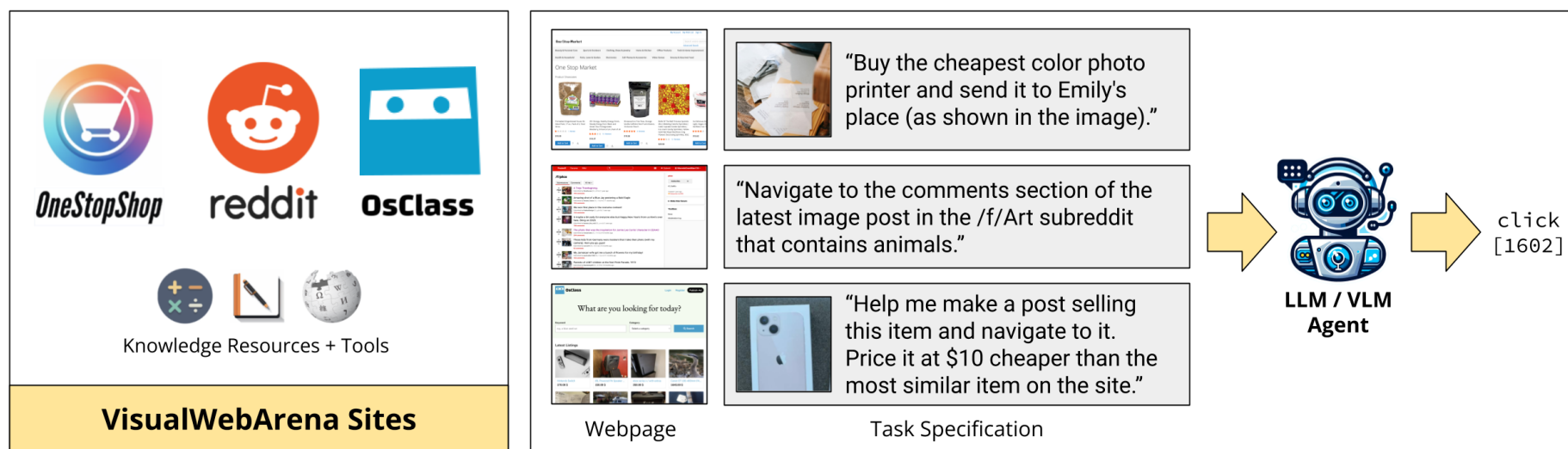
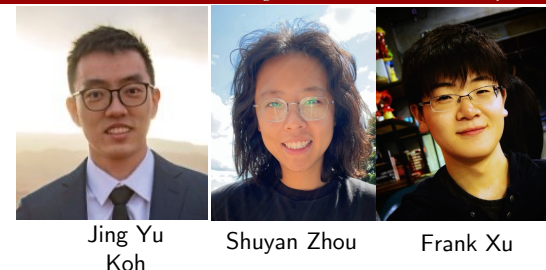

Task: Help me navigate to a shirt that has this on it.



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  /p_som_cot_id_actree_3s.json \  
  --start_url "https://www.amazon.com/" \  
  \  
  --image "https://media.npr.org/assets/  
  img/2023/01/14/this-is-fine_wide-0077dc0  
  607062e15b476fb7f3bd99c5f340af356-s1400-  
  c100.jpg" \  
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VisualWebArena

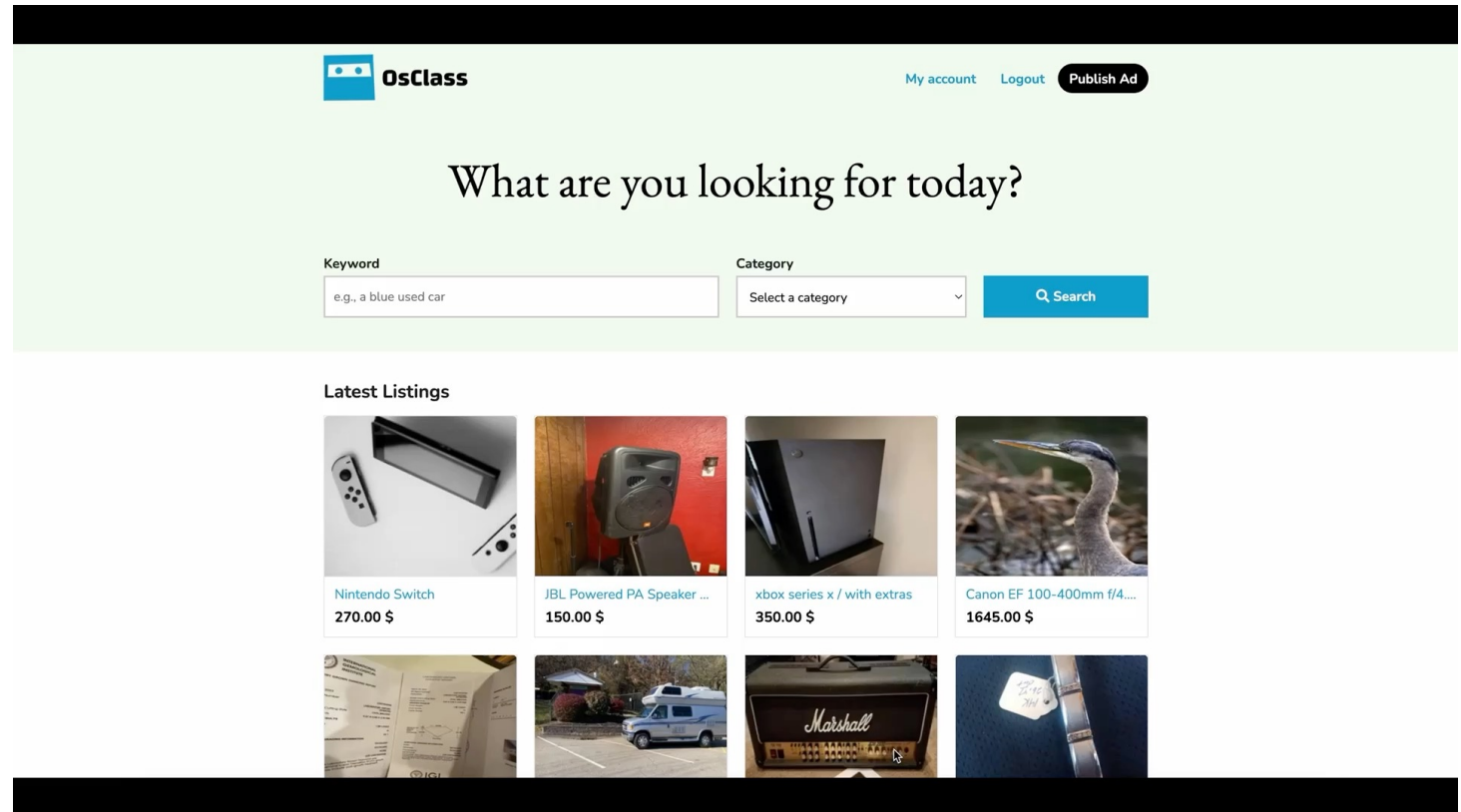
- Build and track the progress of **multimodal agents**
- We design visually grounded tasks to test these abilities
- Visual inputs (and outputs) allow for unique, interesting, and realistic tasks

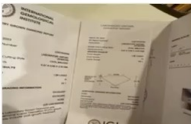


VisualWebArena: Classifields



Task: Find this exact bike that's listed for \$300-500 and post a comment offering \$10 less than their asking price.

The screenshot shows the OsClass website interface. At the top, there's a navigation bar with the OsClass logo, links for "My account", "Logout", and a "Publish Ad" button. Below this is a large heading "What are you looking for today?". Underneath, there are two input fields: "Keyword" with a placeholder "e.g., a blue used car" and "Category" with a dropdown menu labeled "Select a category". A blue "Search" button is to the right. Below the search area, there's a section titled "Latest Listings" displaying a grid of eight items. Each item has a thumbnail image, a title, and a price. The items are: a Nintendo Switch (270.00 \$), a JBL Powered PA Speaker (150.00 \$), an Xbox series x / with extras (350.00 \$), a Canon EF 100-400mm f/4... (1645.00 \$), a document, a white van, a Marshall amplifier, and a metal bracelet.

Latest Listings	
	
	
	
	

VisualWebArena

POMDP environment: $\mathcal{E} = \langle \mathcal{S}, \mathcal{A}, \mathcal{O}, \mathcal{T} \rangle$

- Observations $\mathcal{O}$

The image displays three browser windows from the VisualWebArena environment, illustrating the observation space $\mathcal{O}$.

- Left Window:** A webpage titled "Patio, Lawn & Garden" from "webarena.onestopshop.com". It shows various outdoor furniture items like tables, chairs, and planters.
- Middle Window:** The HTML DOM tree for the webpage. Key elements highlighted include:
 - `<a href="..."></a>`
 - `<div class="..."><a href="...">Outdoor Patio ...</a>`
 - `<div><span>Rating:</span><div><span>82%</span></div></div>`
 - `<a href="...#reviews">12</a>`
- Right Window:** The internal state of the environment, showing the current page title "Patio, Lawn .." and various UI elements like "link 'Image'", "img 'Image'", "link 'Outdoor Patio..'", "LayoutTable", "StaticText 'Rating:'", "generic '82%'", "link '12 Reviews'", "StaticText '\$49.99'", "button 'Add to Cart' focusable: True", "button 'Wish List' focusable: ...", and "button 'Compare' focusable: ...".

- Actions $\mathcal{A}$

Action Type a	Description
click [elem]	Click on element elem.
hover [elem]	Hover on element elem.
type [elem] [text]	Type text on element elem.
press [key_comb]	Press a key combination.
new_tab	Open a new tab.
tab_focus [index]	Focus on the i-th tab.
tab_close	Close current tab.
goto [url]	Open url.
go_back	Click the back button.
go_forward	Click the forward button.
scroll [up down]	Scroll up or down the page.
stop [answer]	End the task with an optional output.

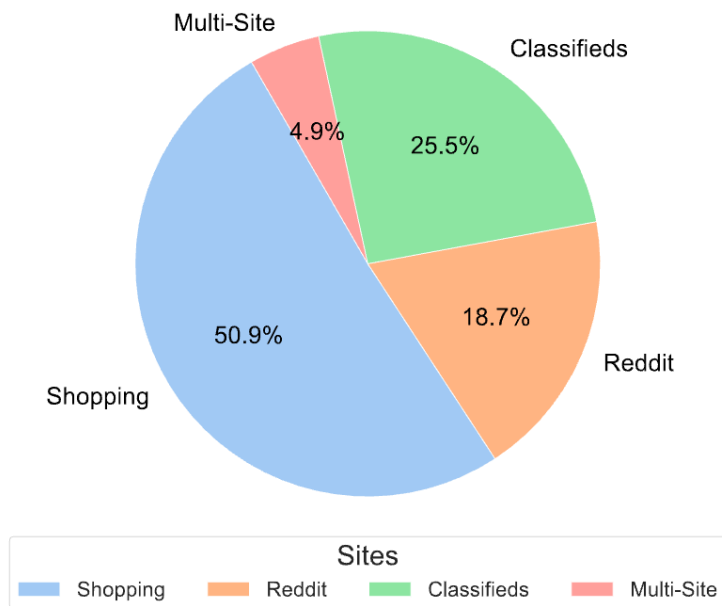
- Deterministic transition function

$$\mathcal{T} : \mathcal{S} \times \mathcal{A} \longrightarrow \mathcal{S}$$

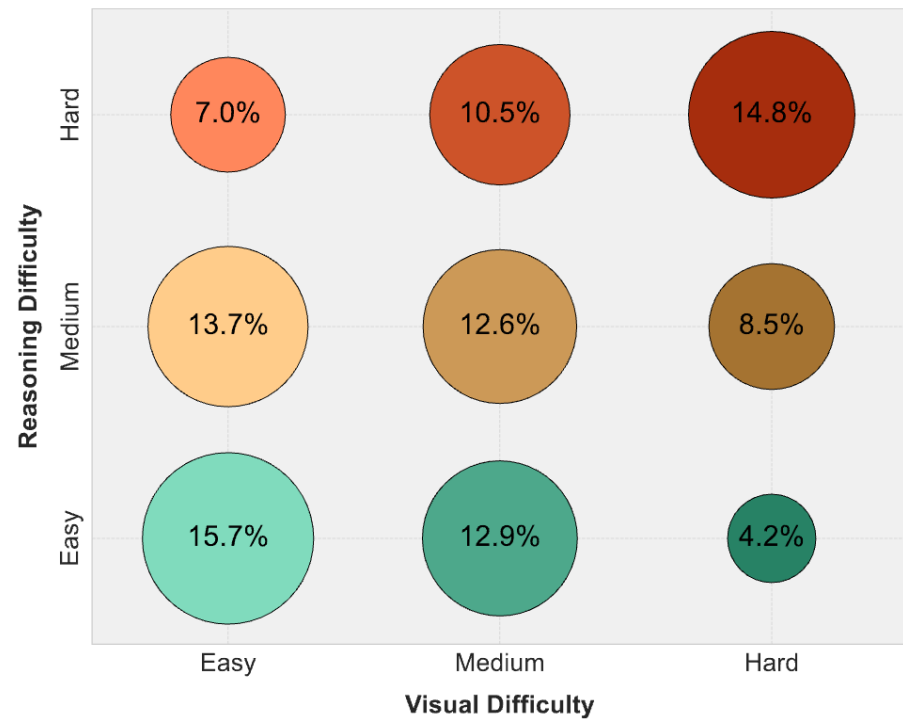
- Reward function: $r(\mathbf{a}, \mathbf{s})$

VisualWebArena

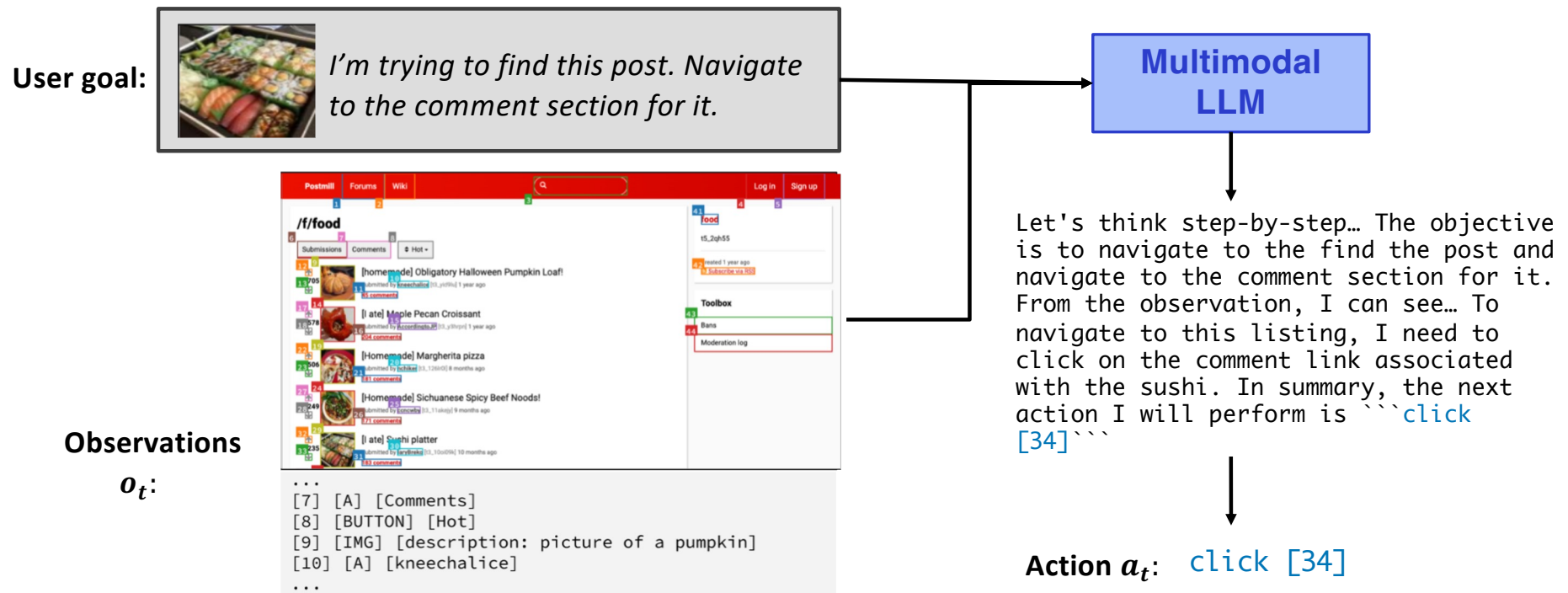
Distribution of Tasks Across Sites



Distribution of Tasks by Difficulty

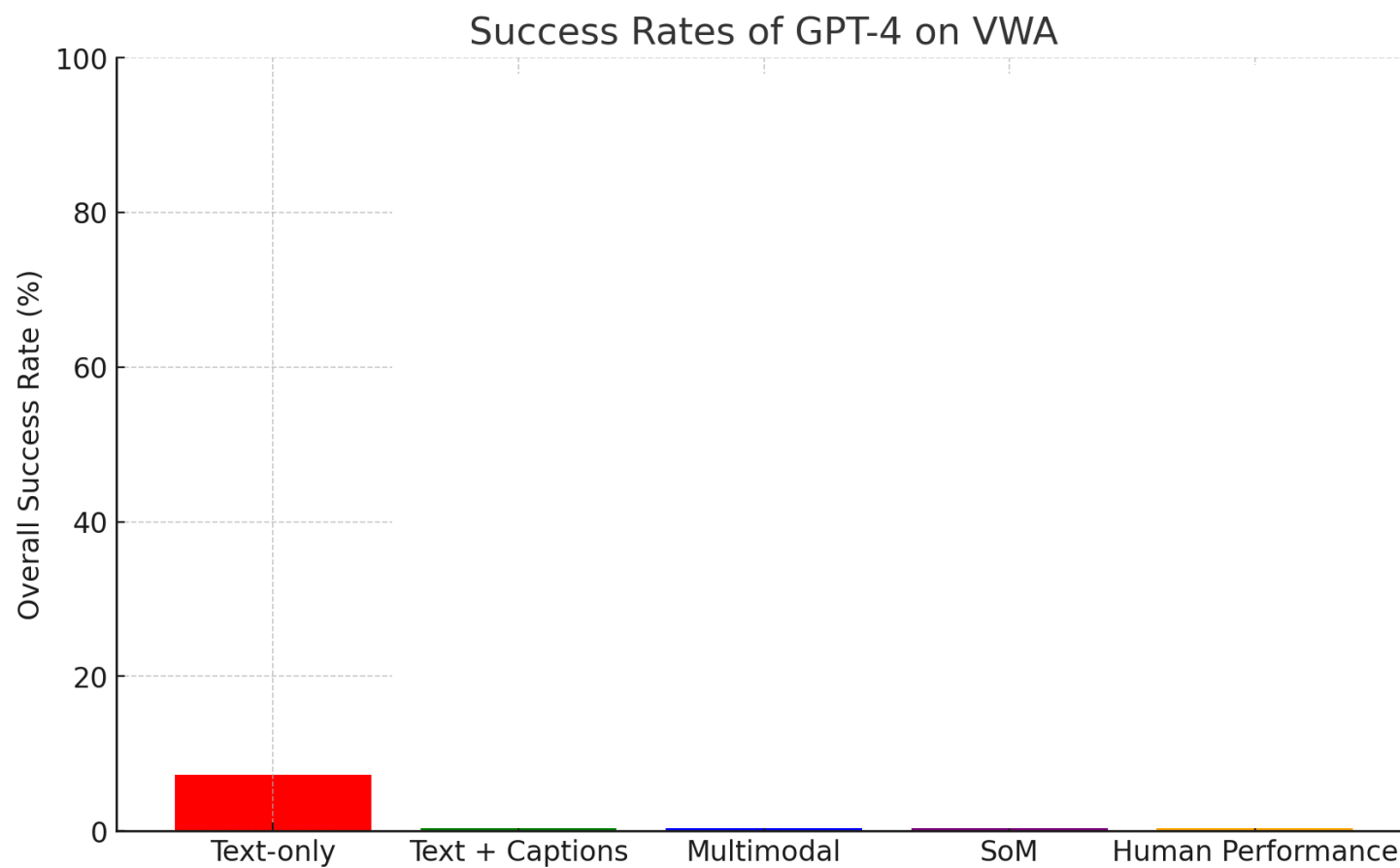


Visual Language Models as Agents



VLM + SoM: Simplified representation with Set-of-Marks (SoM) prompting over interactable elements.

Baseline Agents

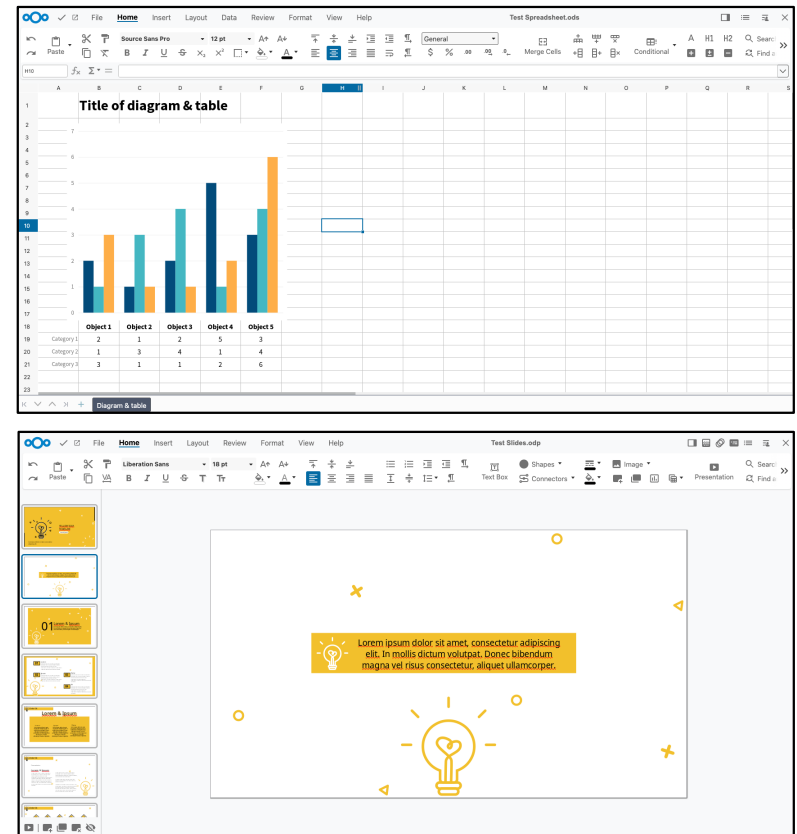


Measuring Productive Tasks

VisualWebArena is a step towards building general purpose agents. But:

- Tasks are not very **consequential**: do not represent significant economic value
- Tasks are simpler, as current LLM agents do not even do well on these problems

Long term: Automate productive, economically valuable tasks

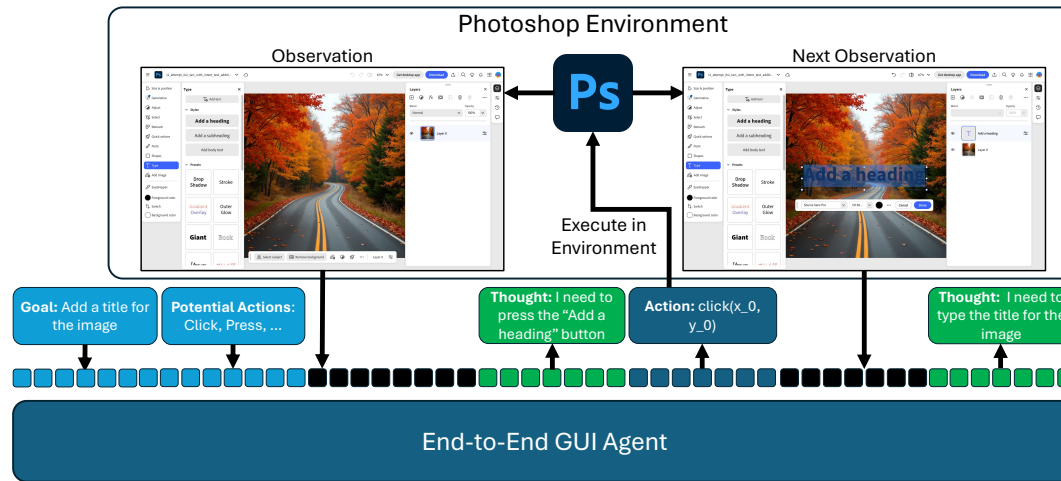


Examples from [Collabora Online](#) / LibreOffice.

Digital Agents for Image Editing Workflows



Maxwell Jones
Rales Fellow



Editing Task: Add a canoe on the lake in the foreground of the image



What is Missing?

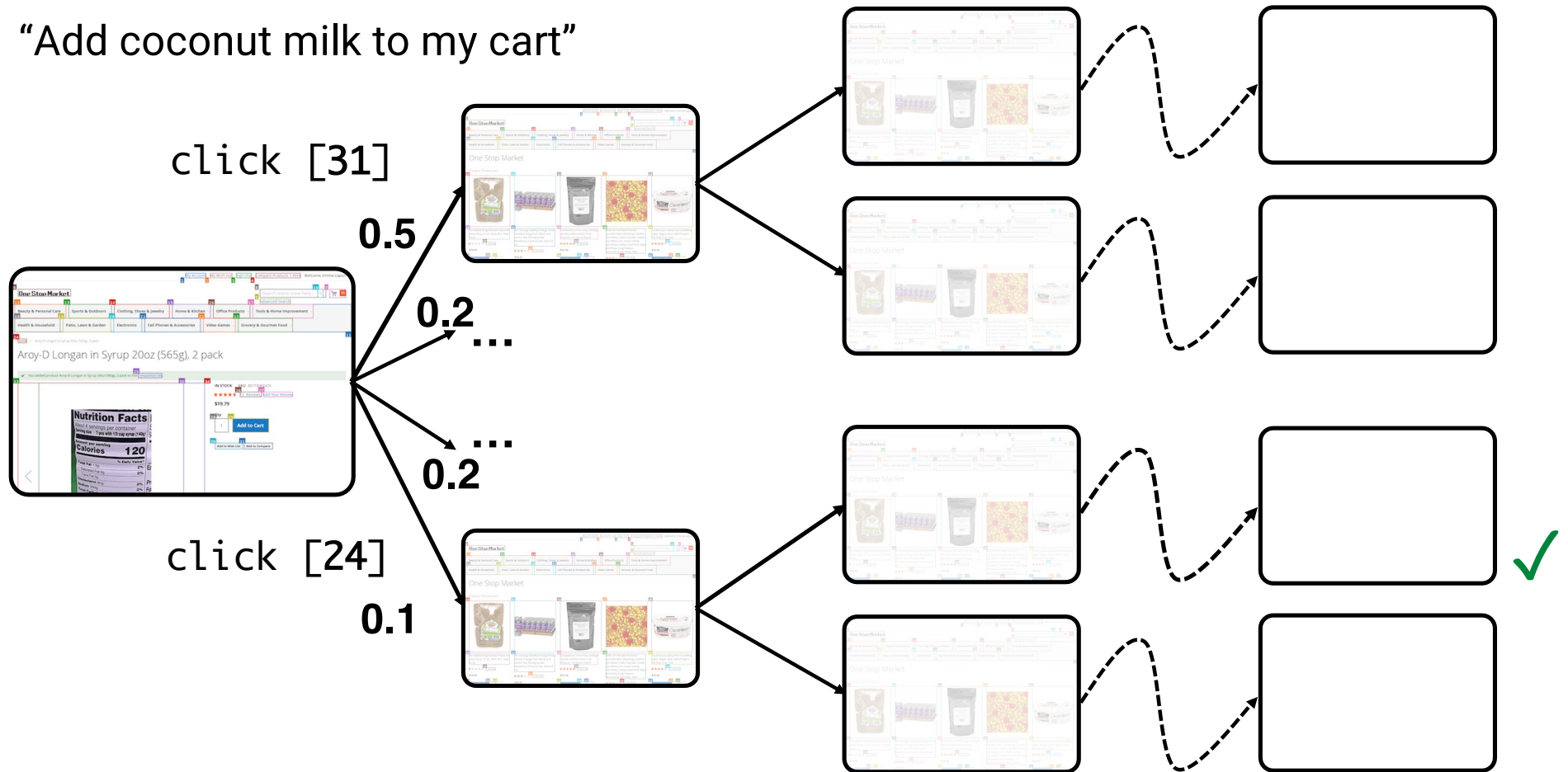
- We need to do a lot more to close the gap:
 - **Reasoning** and **Planning** over long horizons
 - Allow agent to **Search**, execute and coordinate multiple instances in parallel and ask for clarifications/confirmations
 - Strong vision-language-code models
 - Identifying the appropriate level of abstraction for agents (HTML/screenshots/APIs)
- **Multimodal models:** Many real-world tasks require visual grounding to effectively solve (e.g., every task involving PowerPoint, Excel, Photoshop). To develop strong general agents, we will need to train and build strong vision-language models.

Exponential Error Compounding in Agents

Accuracy @ k steps:				
1 (single step)	5	10	30	50
90%	59.05%	34.87%	4.24%	0.52%
95%	77.38%	59.87%	21.46%	7.69%
99%	95.10%	90.44%	73.97%	60.50%
99.9%	99.50%	99.00%	97.04%	95.12%
99.99%	99.95%	99.90%	99.70%	99.50%

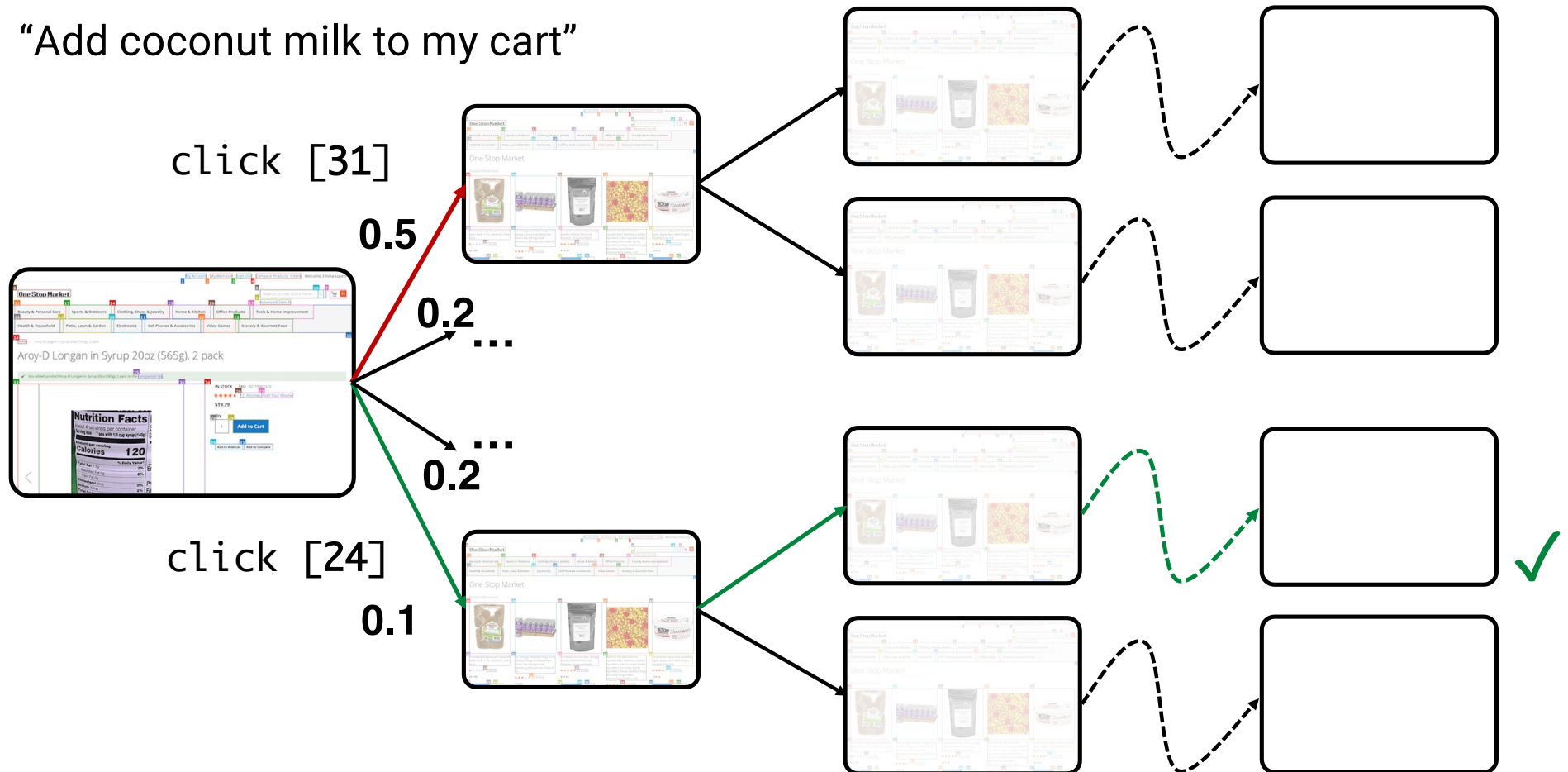
Local Decisions; Global Consequences

“Add coconut milk to my cart”



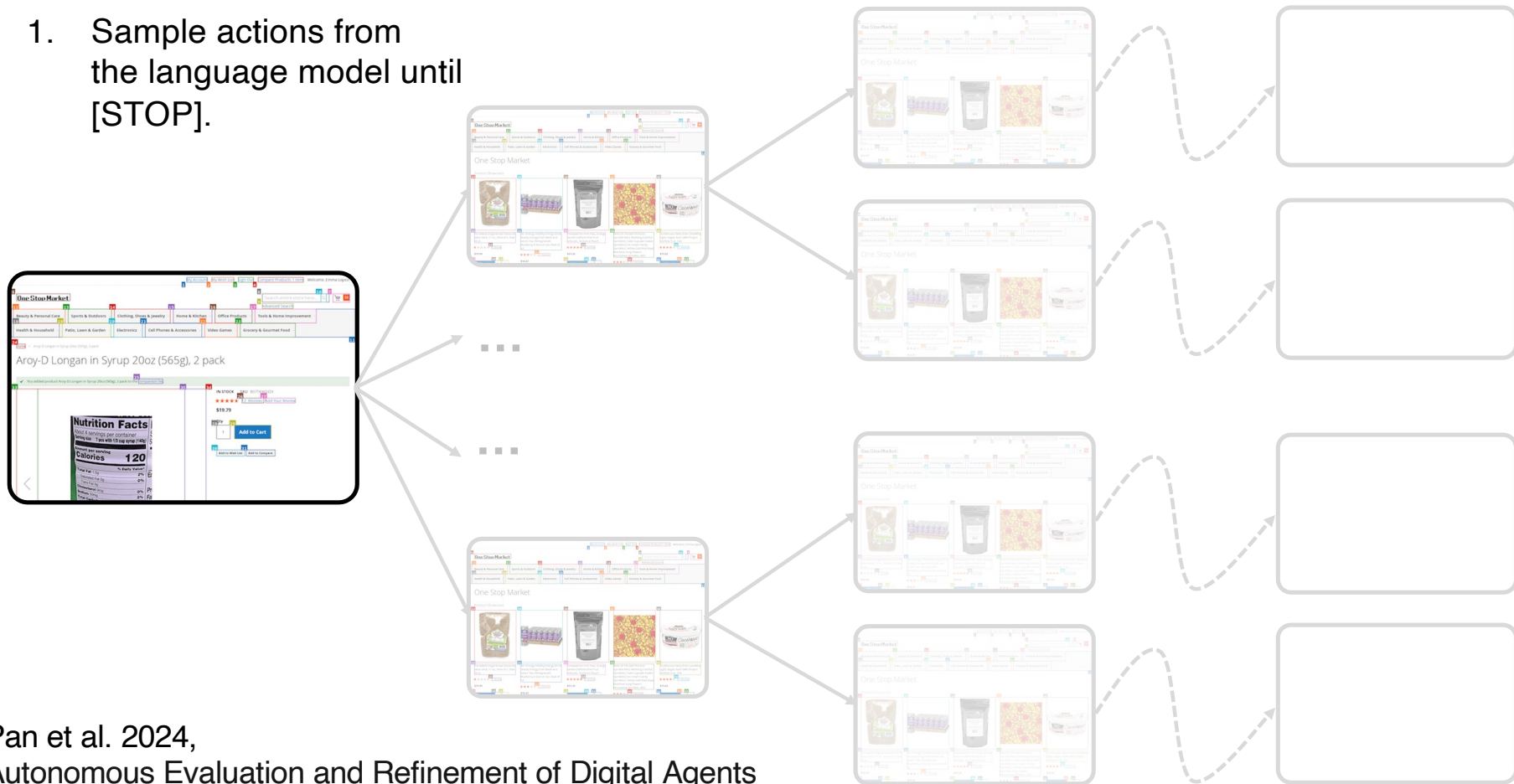
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Search By Repeated Sampling

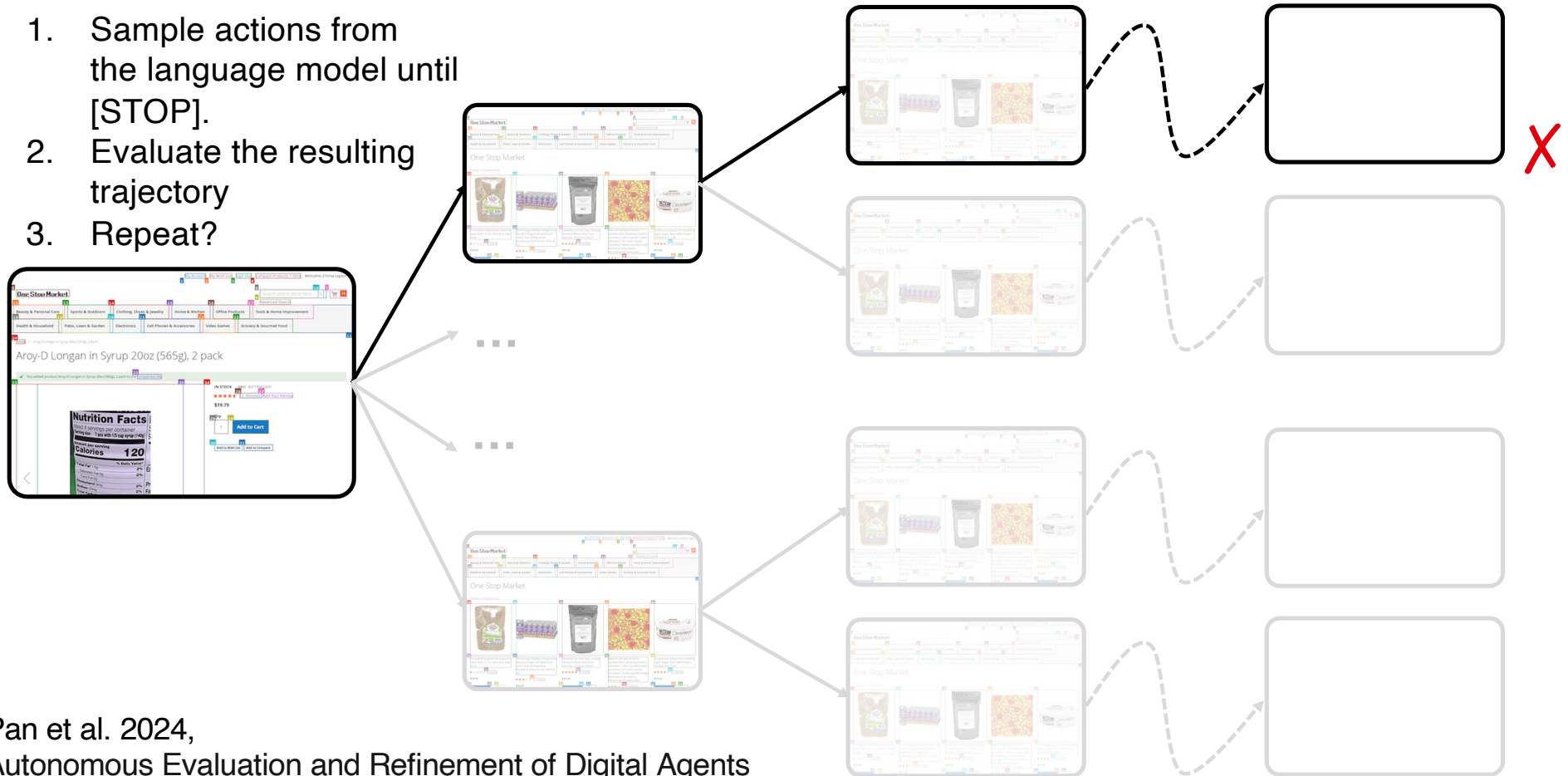
1. Sample actions from the language model until [STOP].



Pan et al. 2024,
Autonomous Evaluation and Refinement of Digital Agents

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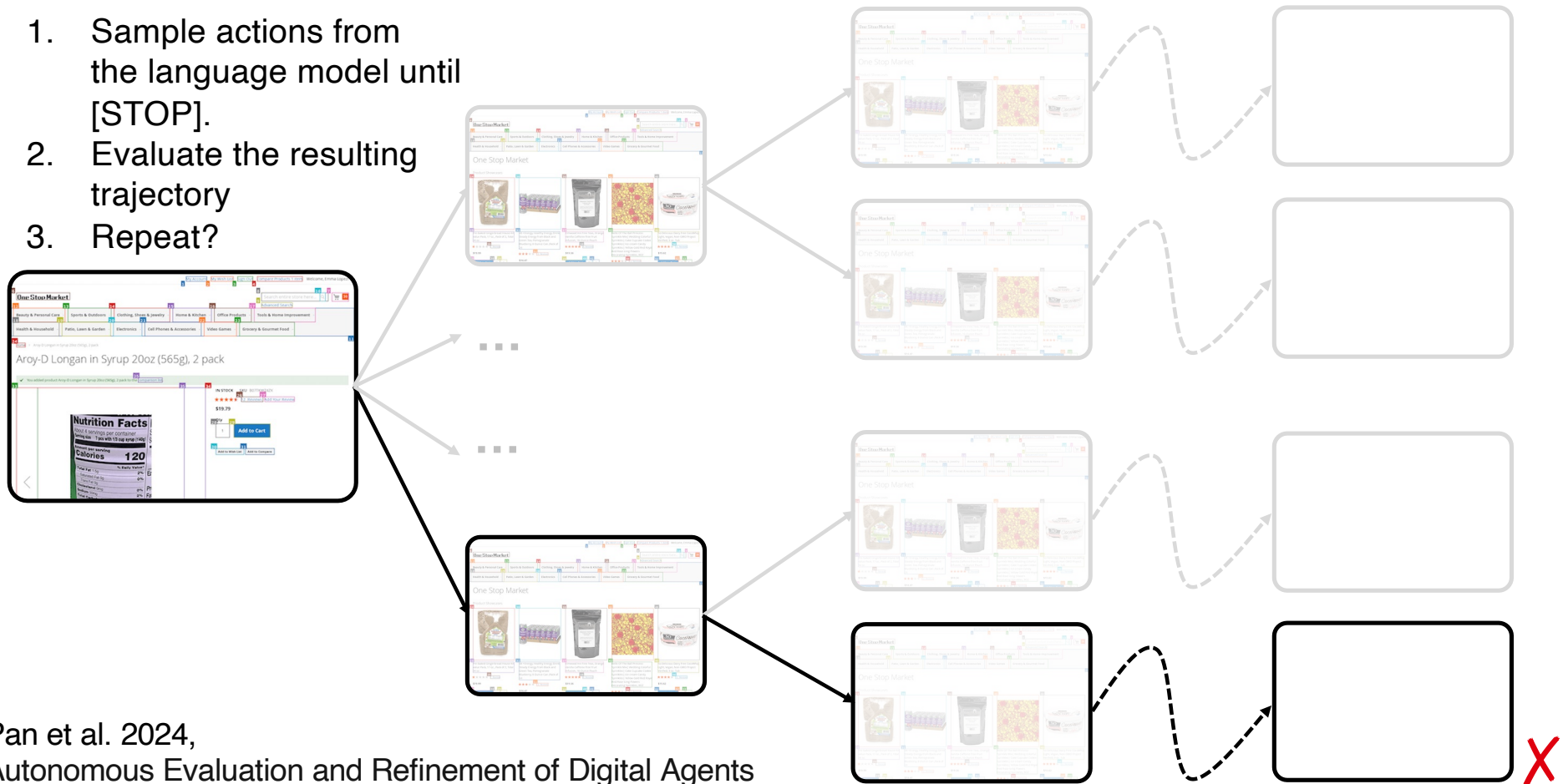
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2. Evaluate the resulting trajectory
3. Repeat?



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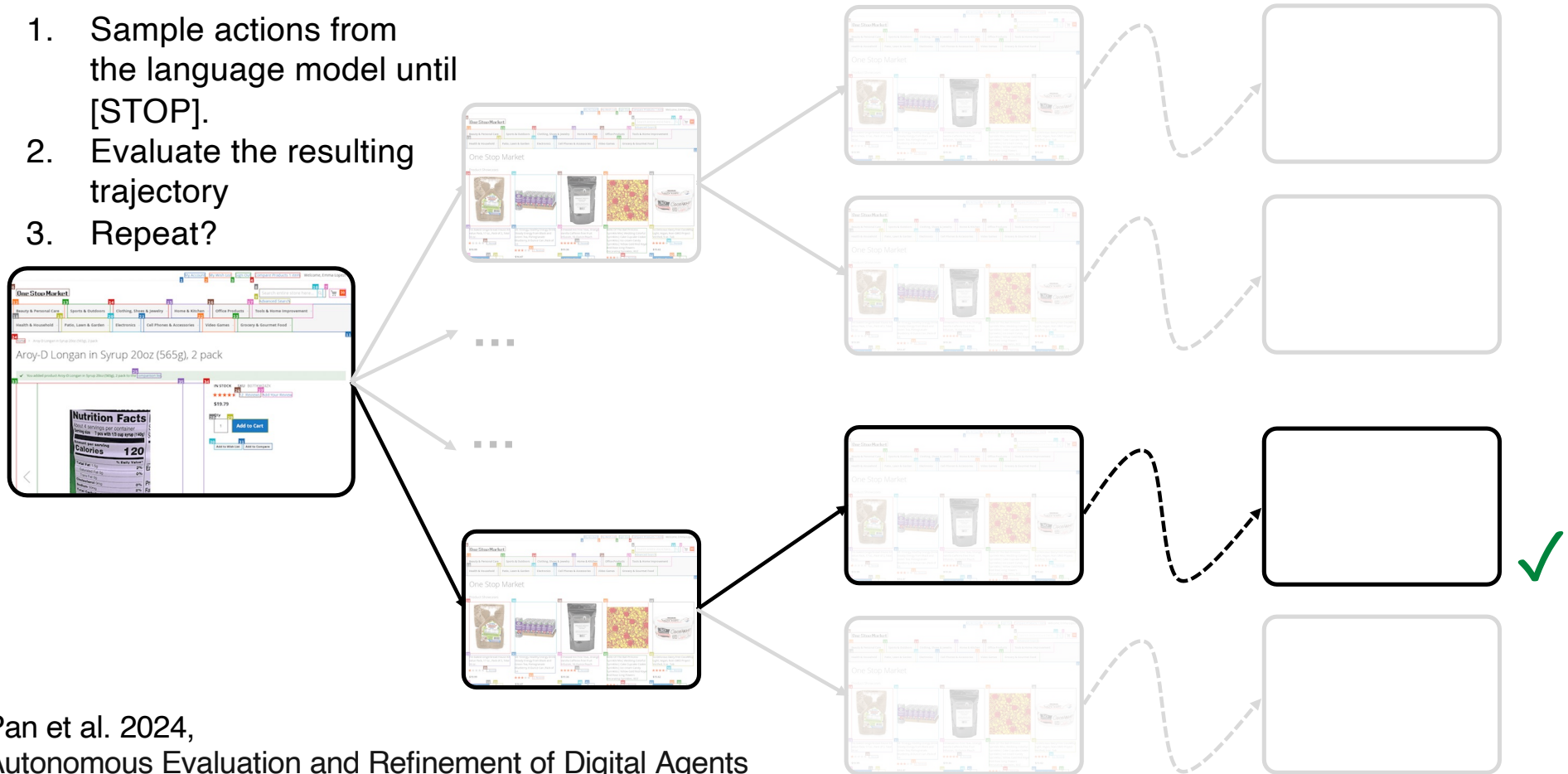
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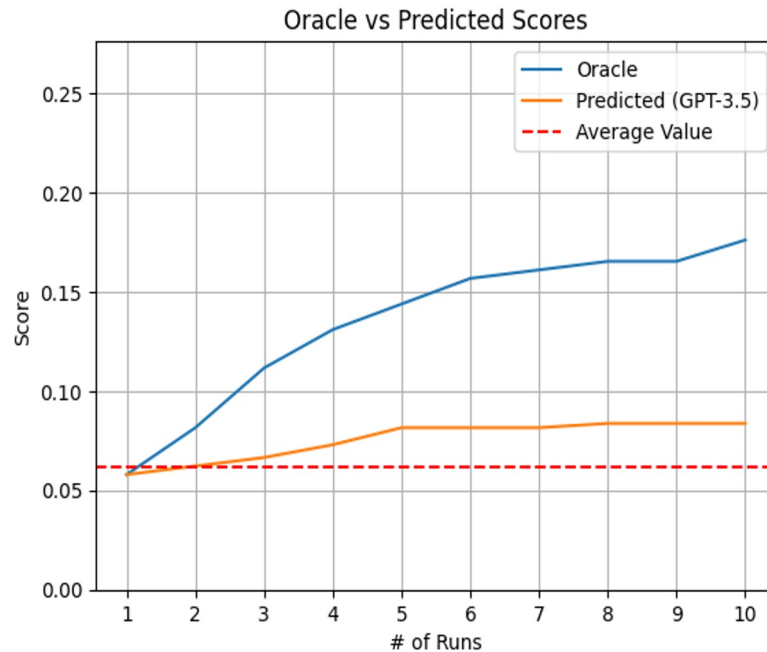
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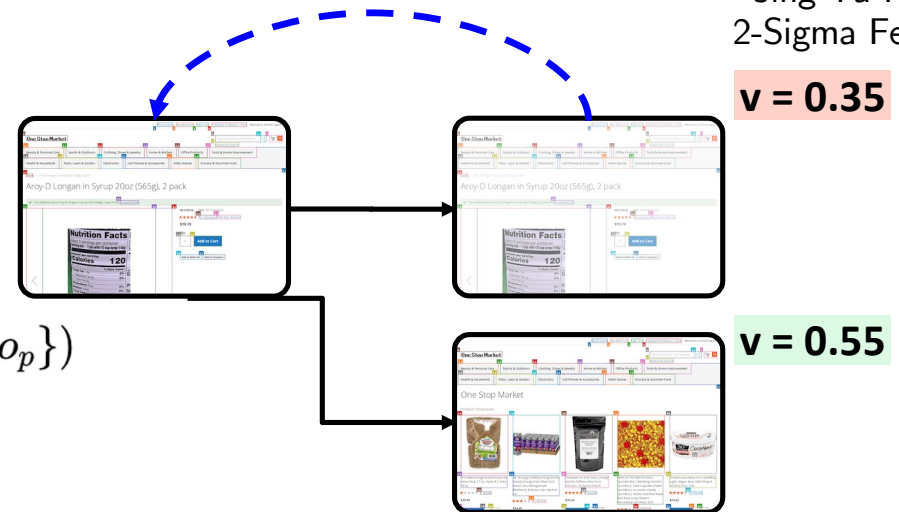
Repeated sampling helps!

- But the space is exponentially large. Can we guide exploration?
- **Key idea:** apply value function to intermediate nodes.

Our Method: Tree Search

- Best-first search algorithm
- Ingredients:
 - Baseline agent to propose actions.
 - Way to backtrack in the environment.
 - A **value function** $v_p = f_v(I, \{o_1, \dots, o_p\})$ to score and rerank candidate states.

In this work, we prompt a multimodal LLM (GPT-4o) to act as an evaluator.



Jing Yu Koh
2-Sigma Fellow

$v = 0.35$

$v = 0.55$



Task Instruction (I): “Can you add this and the other canned fruit (of the same brand) that looks like this, but red instead of brown to the comparison page?”

Legend

- ① Step sequence
- v = 1.0 State values
- ▶ Backtracking

GPT-4o Agent



GPT-4o Agent + Search



Starting State



Task Instruction (I): “Can you add this and the other canned fruit (of the same brand) that looks like this, but red instead of brown to the comparison page?”

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GPT-4o Agent



GPT-4o Agent + Search



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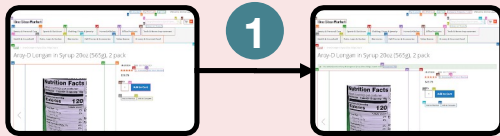


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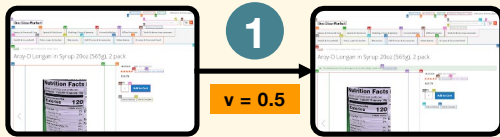
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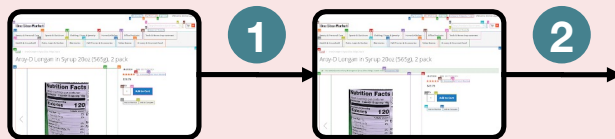


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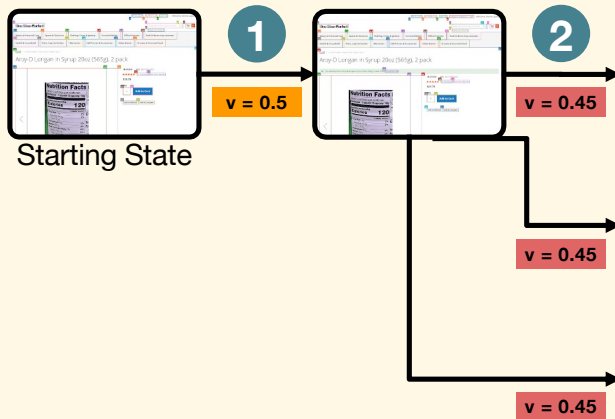
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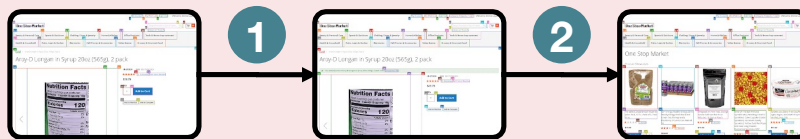


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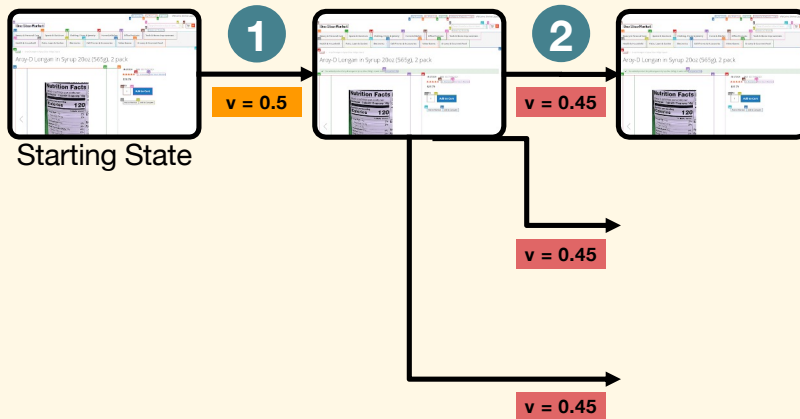
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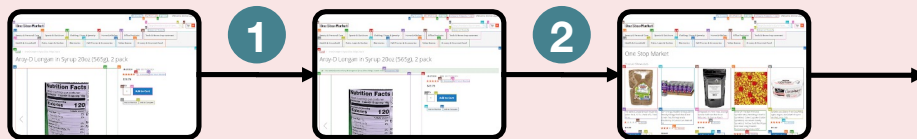


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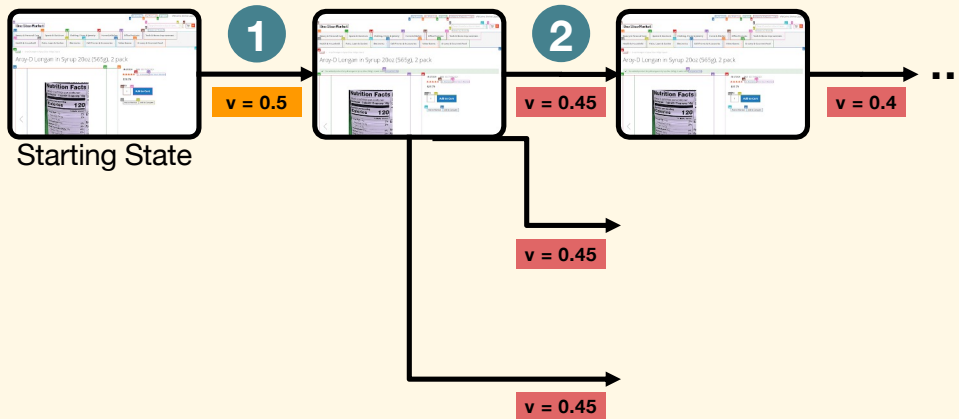
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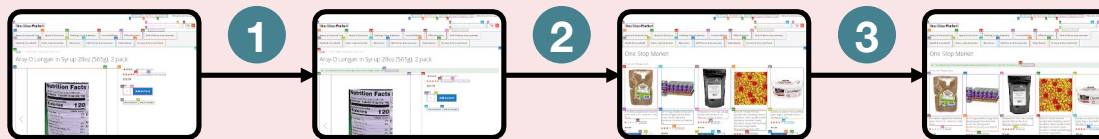


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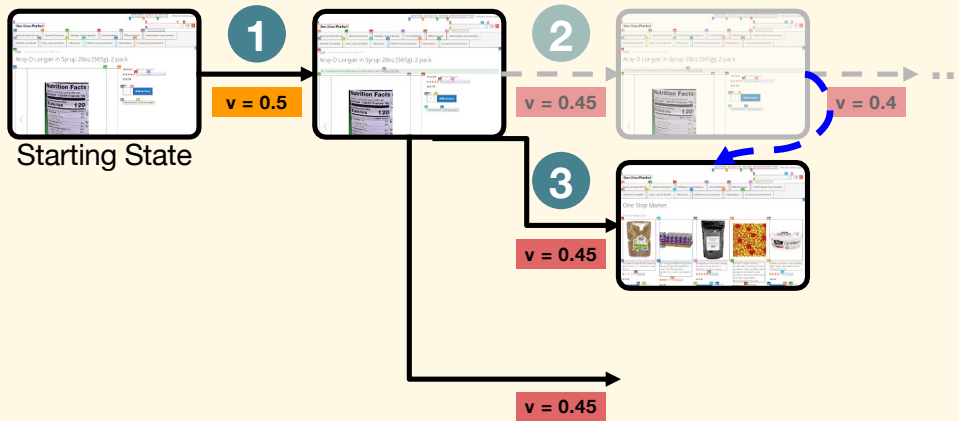
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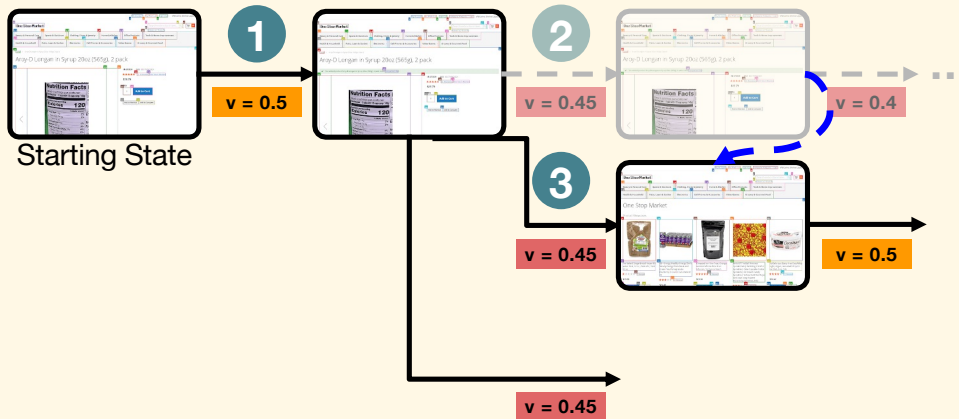
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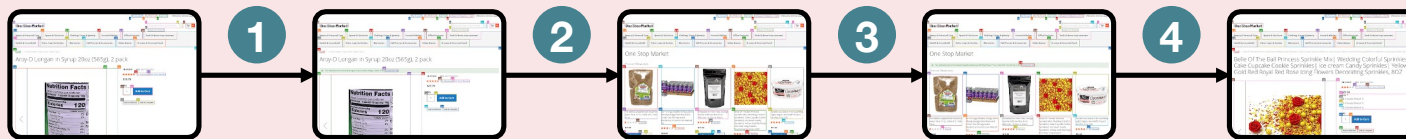


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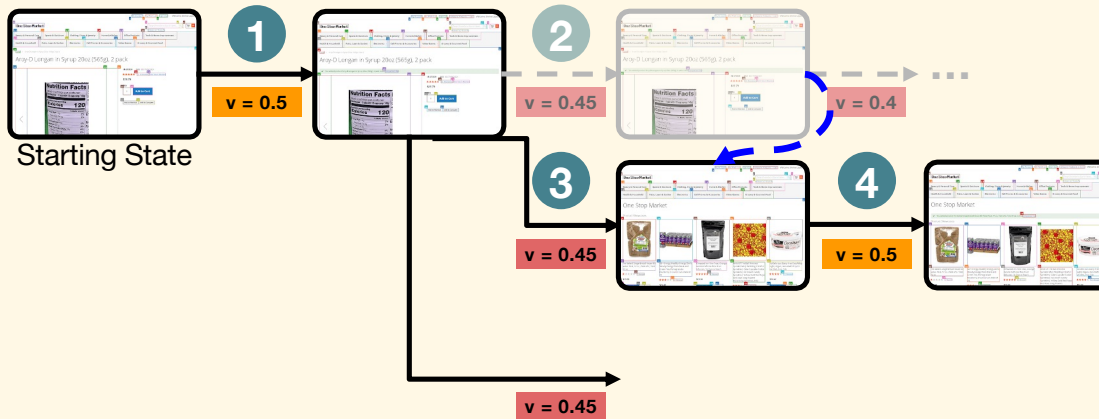
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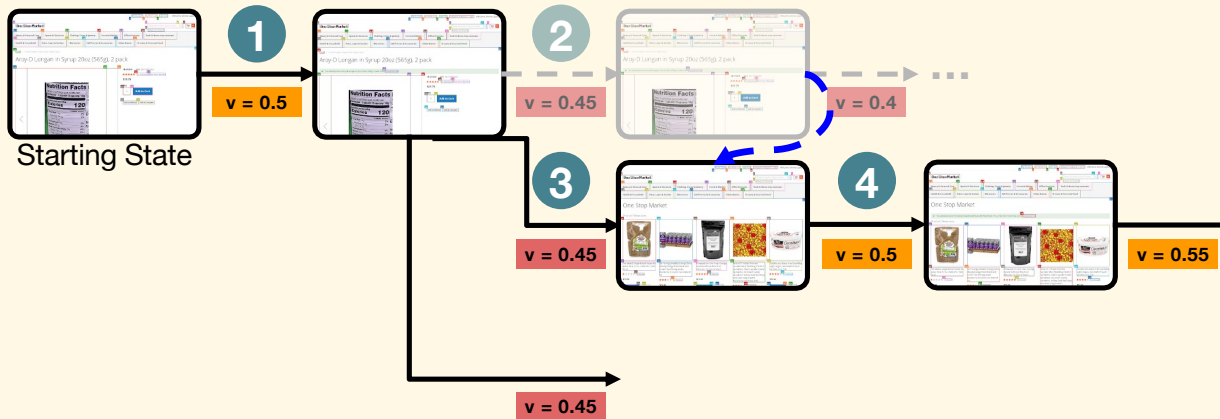
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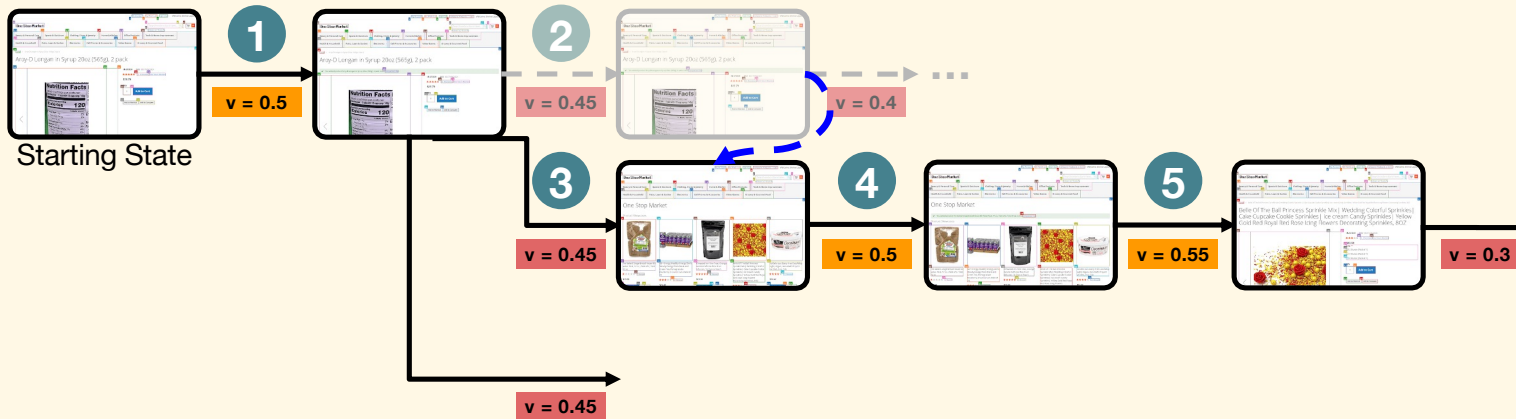
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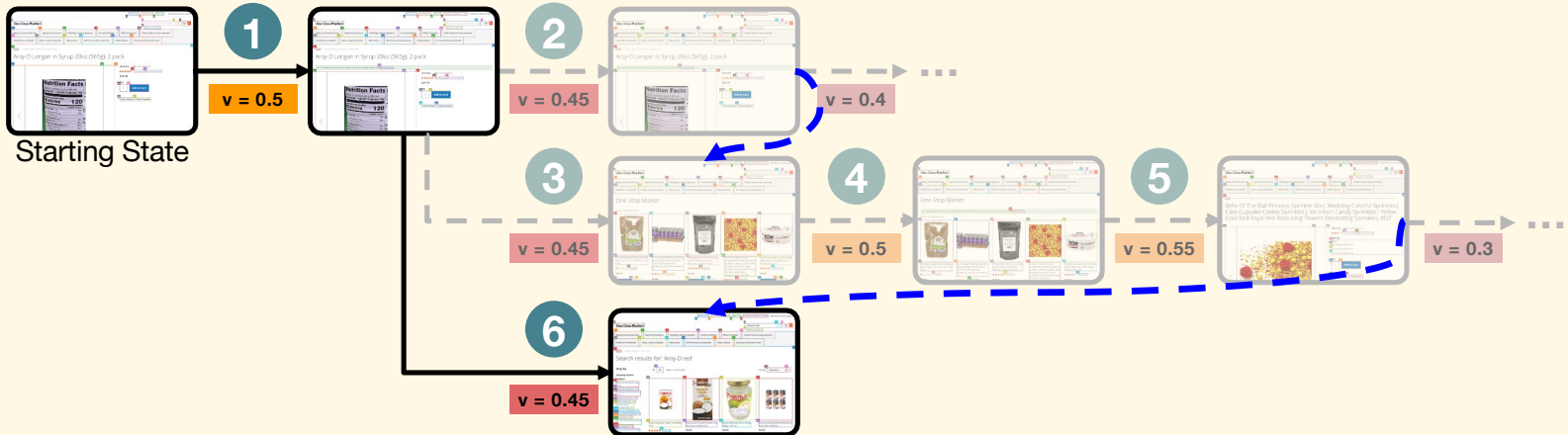
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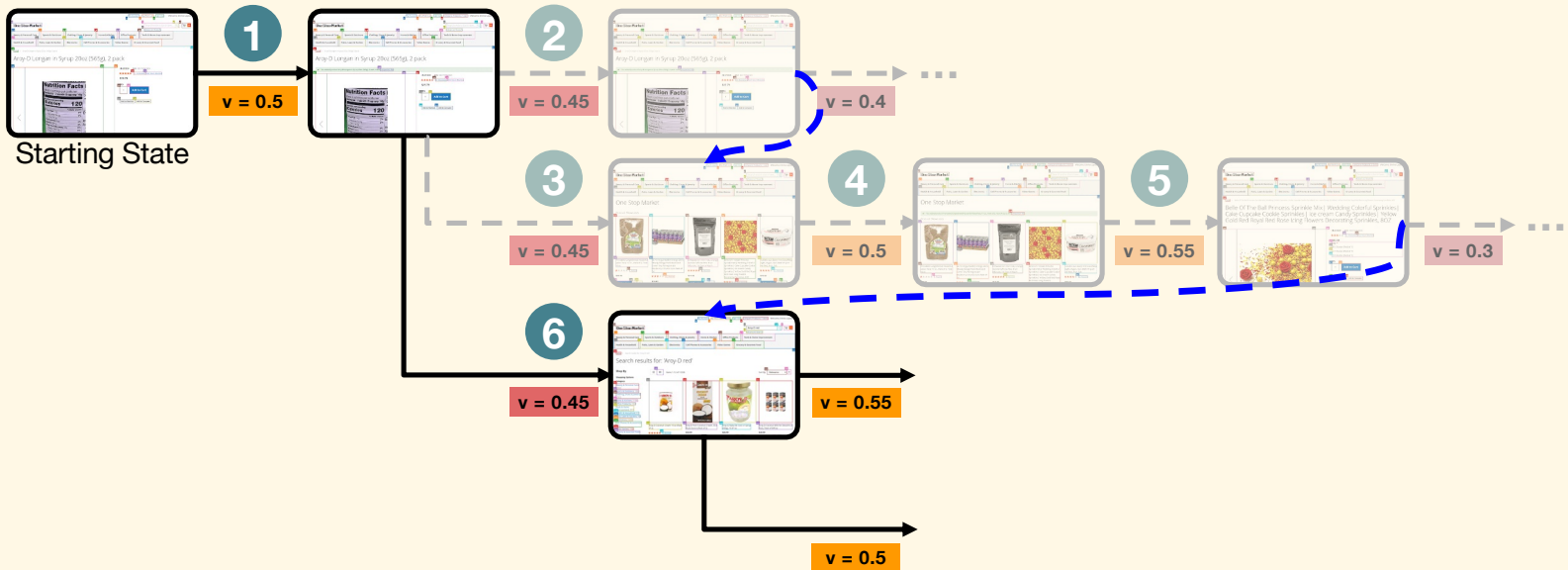
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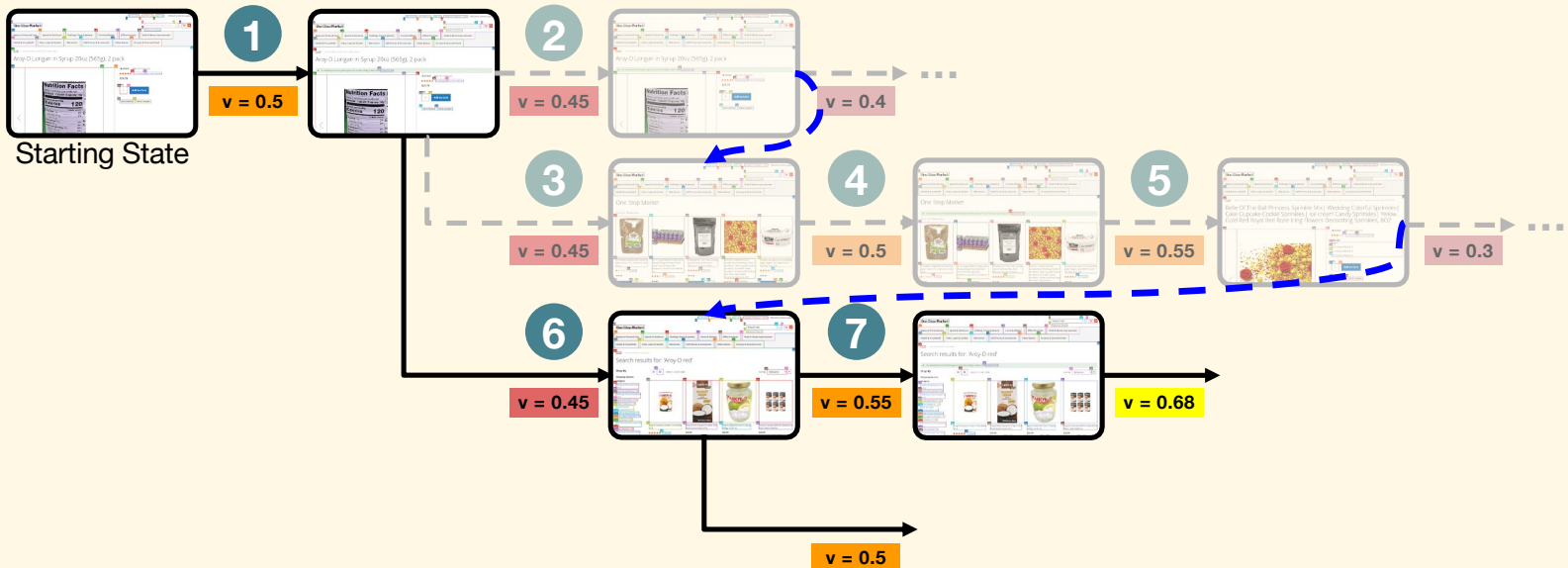
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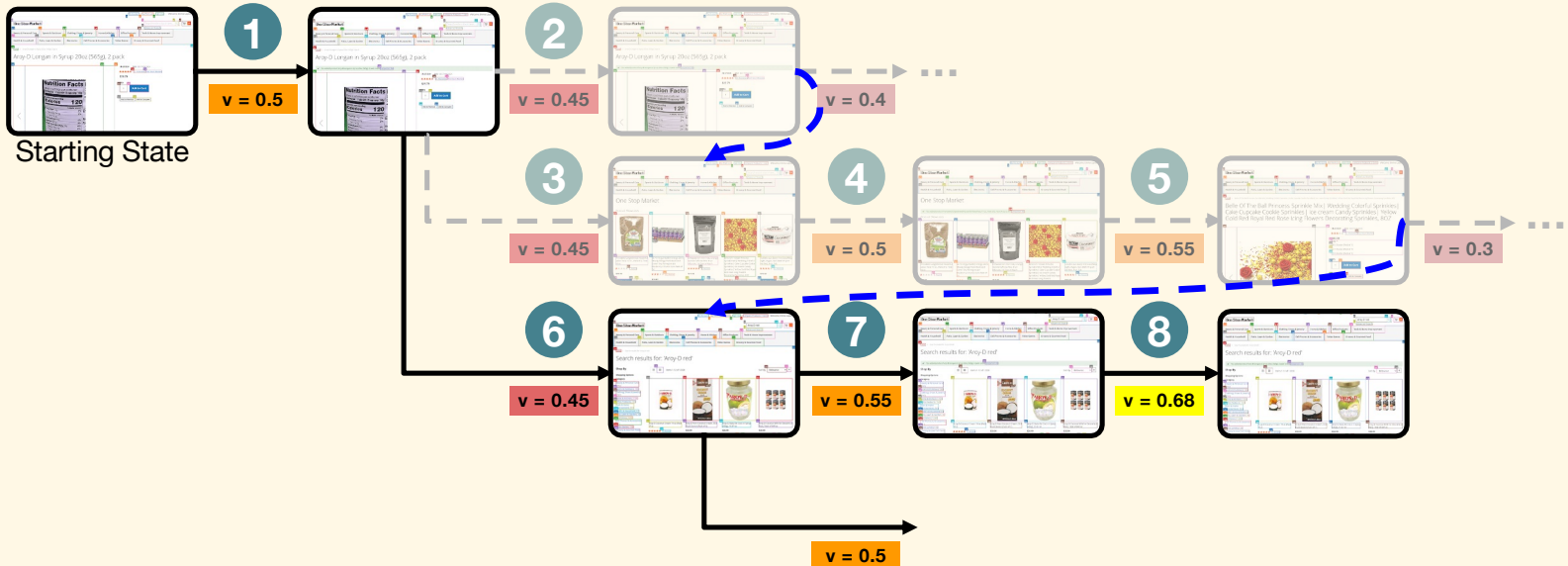
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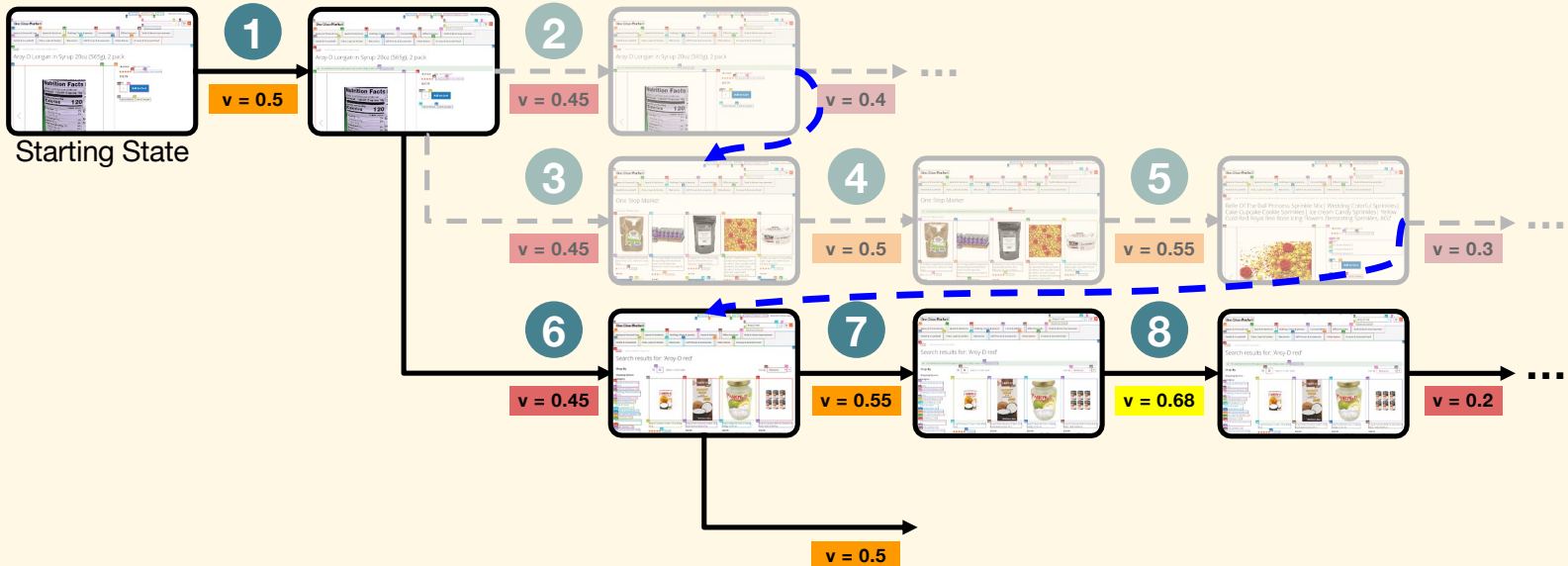
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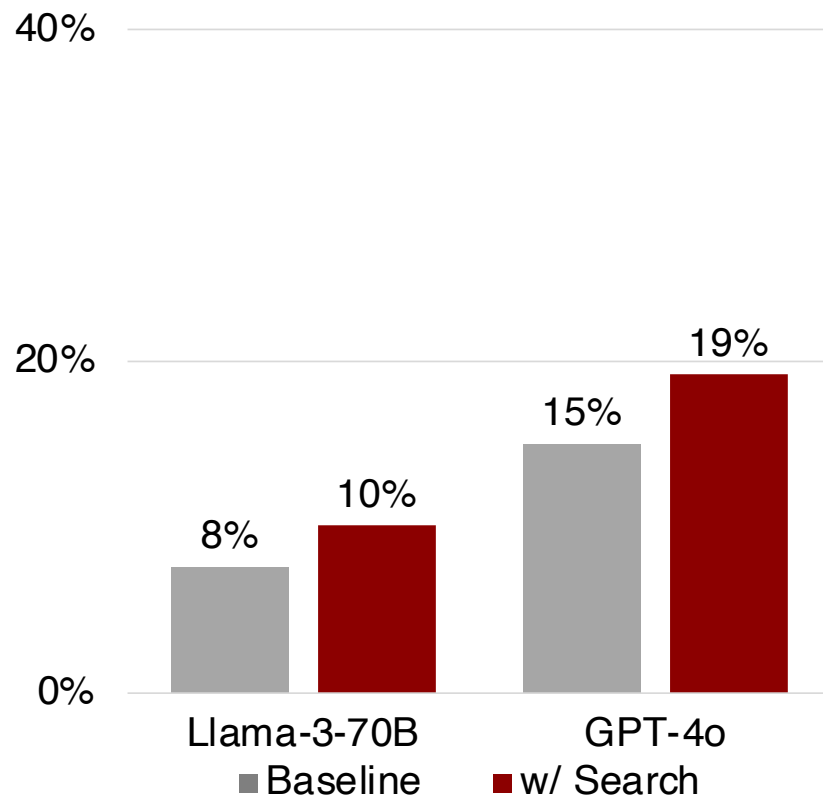


GPT-4o Agent + Search

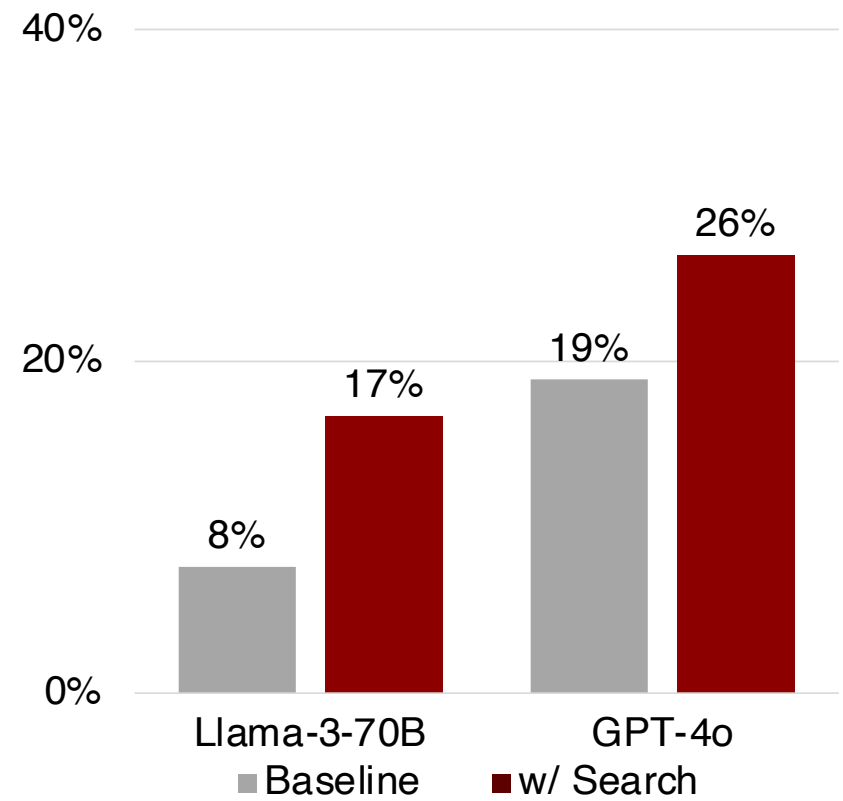


Results

WebArena



VisualWebArena



Current Work

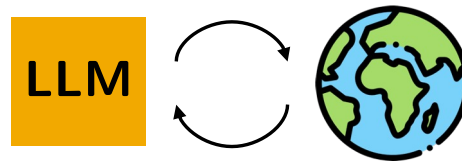
- Search as a policy improvement function
- Improving Value Function by fine-tuning instead of prompting
- Explore compute tradeoff between improving baseline agent vs. doing **more search at inference time**
- What if we don't have a perfect simulator – **how can we collect data at scale?**

Agents Suffer From A Data Problem

- Top LLMs fall short of humans by 68.92% on Visual Web Arena
- LLMs are often **trained offline**, then **deployed zero-shot** as agents

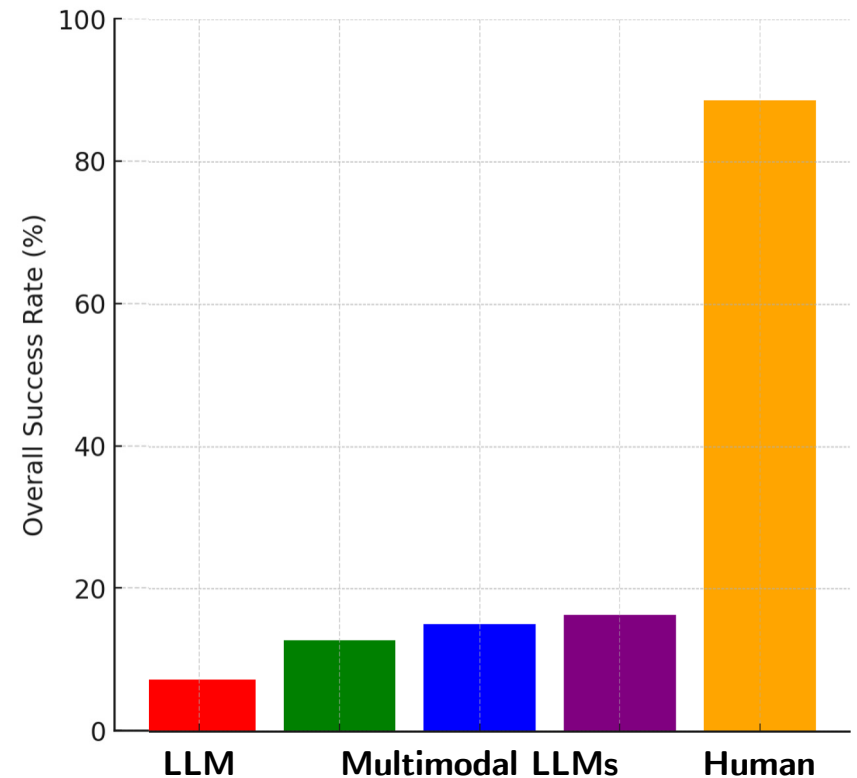


Training Data



Test Environment

Success Rates of GPT-4 on VWA

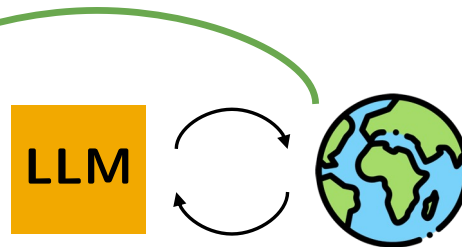


Agents Suffer From A Data Problem

- Top LLMs fall short of humans by 68.92% on Visual Web Arena
- Can **synthetic tasks** unlock internet-scale training for agents?

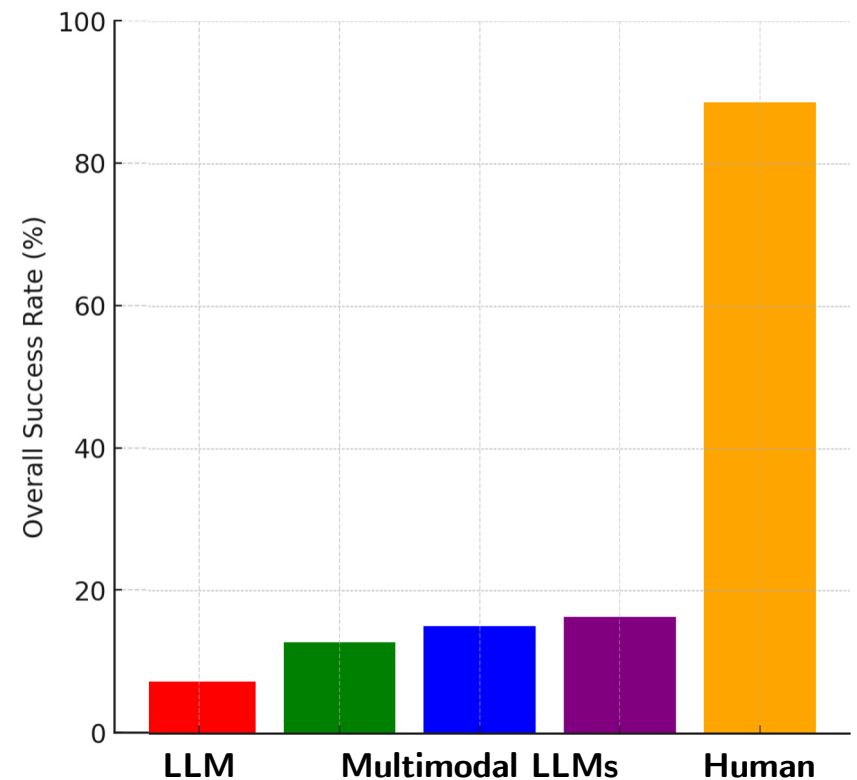


Training Data



Test Environment

Success Rates of GPT-4 on VWA



Find the opening hours
for La Sagrada Familia.

Towards Internet-Scale Training For Agents

- **Goal: unlock internet-scale data for training for agent models.**



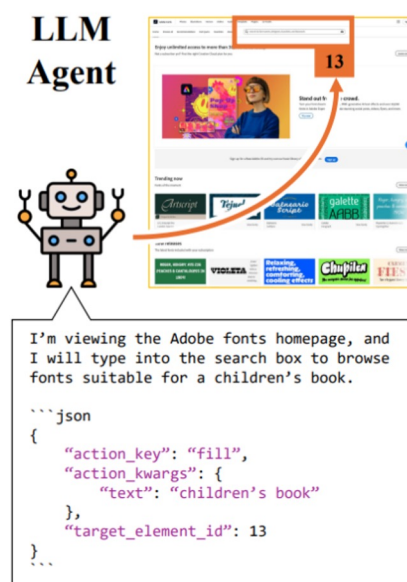
Brandon Trabucco
NDSEG Fellow

Towards Internet-Scale Training For Agents

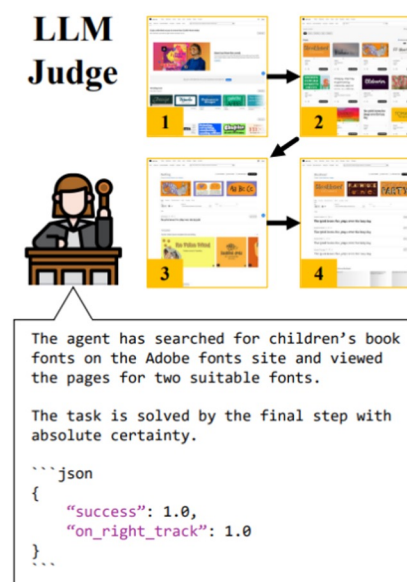
- **Goal: unlock internet-scale data for training for agent models.**
- **Key Idea: LLMs can generate & verify agent tasks on live websites**



Brandon Trabucco
NDSEG Fellow



Stage 2: Attempt



Stage 3: Evaluate

Use Llama To Generate Agentic Tasks

- Given a web domain as text (i.e. merseyferries.co.uk)
 - Propose a realistic task that an average user could complete in one session.
-

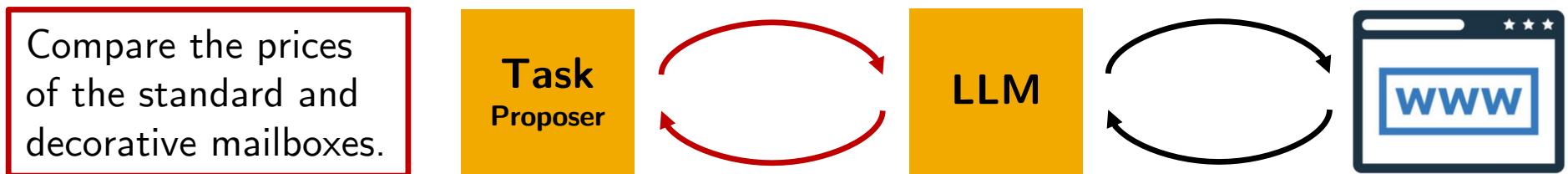
Llama-3.1-70B-Instruct: describe the “merseyferries.co.uk” site

Mersey Ferries (merseyferries.co.uk) is the **official website for the famous Mersey Ferry service** in Liverpool, England. This website serves as a one-stop platform for users to access various services and information related to the ferry operation. Here are some things a user can do on this site:

1. **Plan a Journey:** Users can plan their journey by checking the ferry timetable, which includes departure and arrival times for both commuter and cruise services.
2. **Book Tickets:** Users can book tickets online for both commuter and cruise services, including the popular Manchester Ship Canal Cruise and the Liverpool Bay Cruise.

A Flywheel For Generating Challenging Tasks

- 1) **First, deeply explore** the website (ie: `budgetmailboxes.com`) with an easy task.
- 2) Then, review trajectories, and generate a **challenging task** that **an expert user** would leverage the website to complete.



53

Agent Trajectory

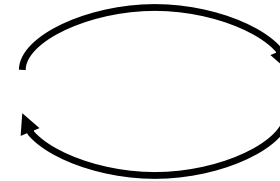
A Flywheel For Generating Challenging Tasks

- 1) **First, deeply explore** the website (ie: `budgetmailboxes.com`) with an easy task.
- 2) Then, review trajectories, and generate a **challenging task** that **an expert user** would leverage the website to complete.

Find the price of the Florence 16 Door Cluster Mailbox CBU with 2 Parcel Lockers, Decorative Crown and Pedestal Included and compare it to the price of the 16 Door USPS Cluster Mailbox CBU with 2 Parcel Lockers, Pedestal Included



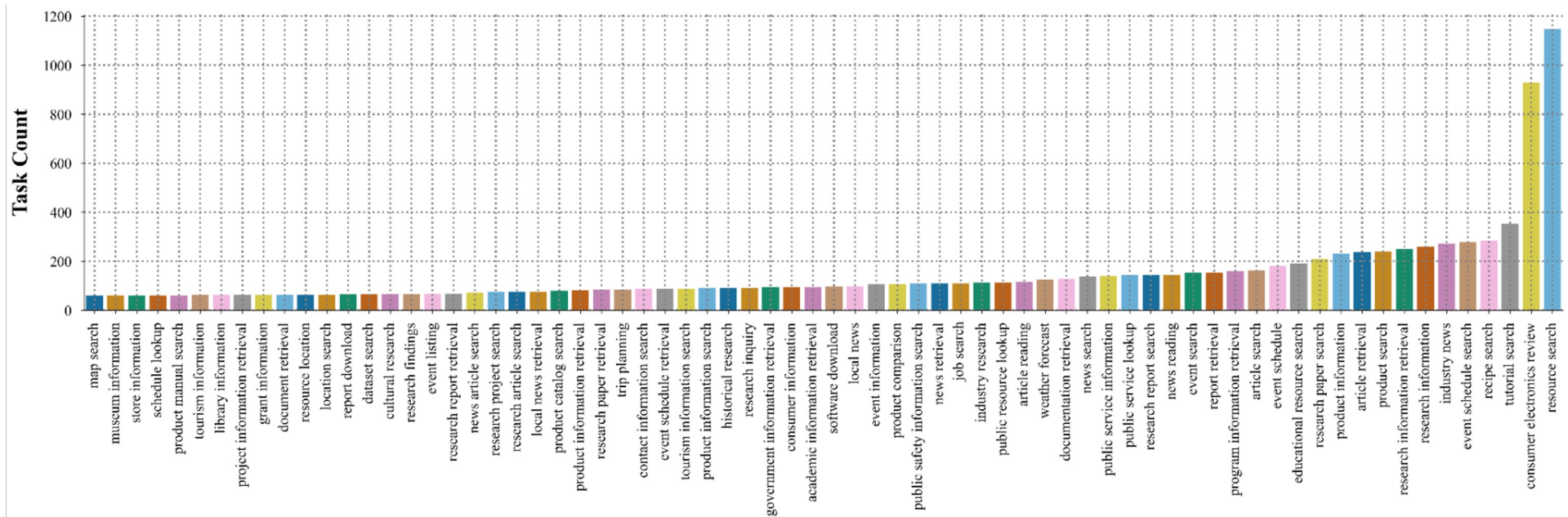
Task Flywheel



Agent Trajectory



Use Llama To Generate Agentic Tasks



- Heavy-tailed distribution, most tasks involve information retrieval
- Crucially, **tasks do not modify the state of the internet**

LLMs Are Competitive With Human Annotators

- We compare the Task Proposer to human annotators at **detecting harmful sites**, and creating **feasible agent tasks**, for a set of 100 examples.

Method	Acc.	Prec.	Recall
<i>Llama 3.1 70B</i>	85%	0.77	1.00
<i>GPT-4o</i>	95%	0.91	1.00
<i>Gemini 1.5 Pro</i>	97%	0.96	0.98

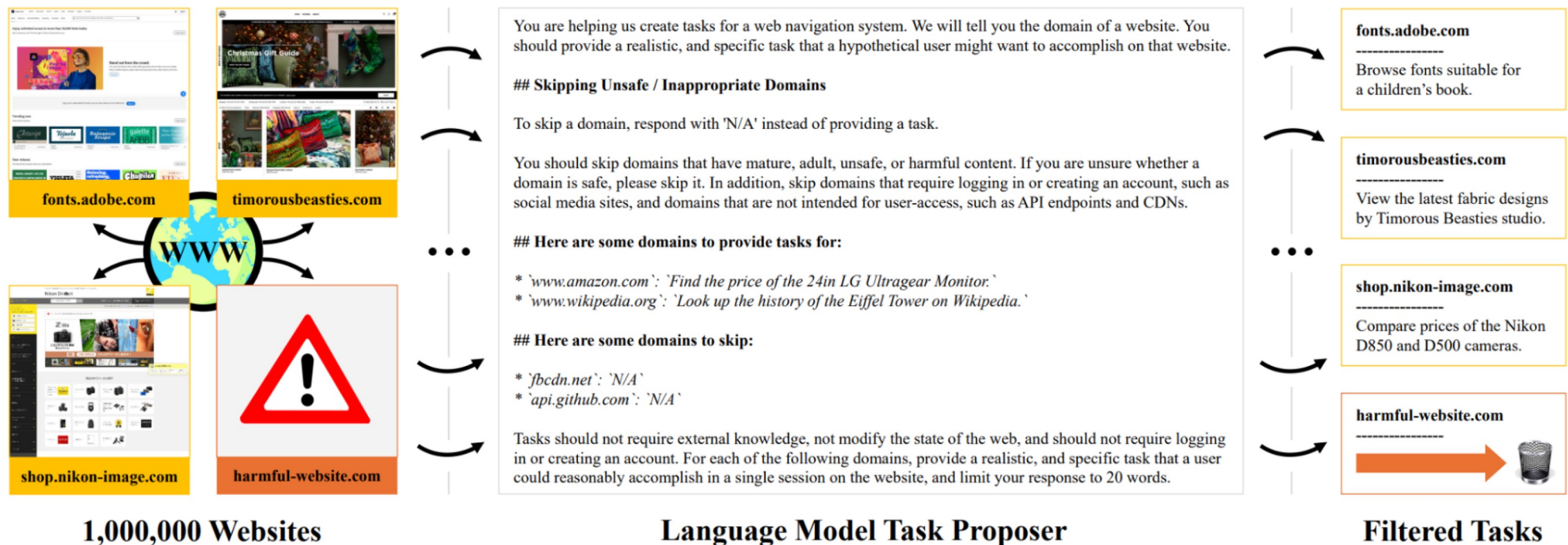
Accuracy For Detecting
Harmful Sites

Method	Feasibility Rate
<i>Llama 3.1 70B</i>	75%
<i>GPT-4o</i>	85%
<i>Gemini 1.5 Pro</i>	89%

Expert Feasibility Of
Proposed Tasks

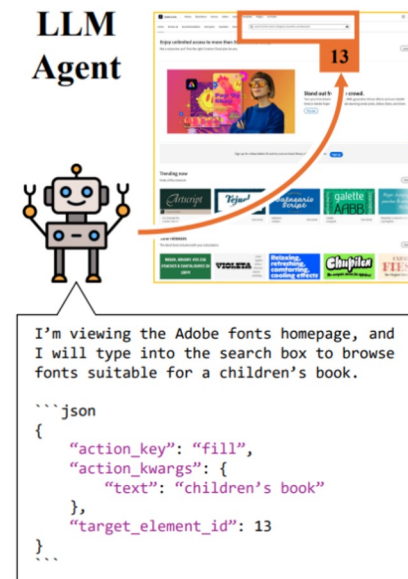
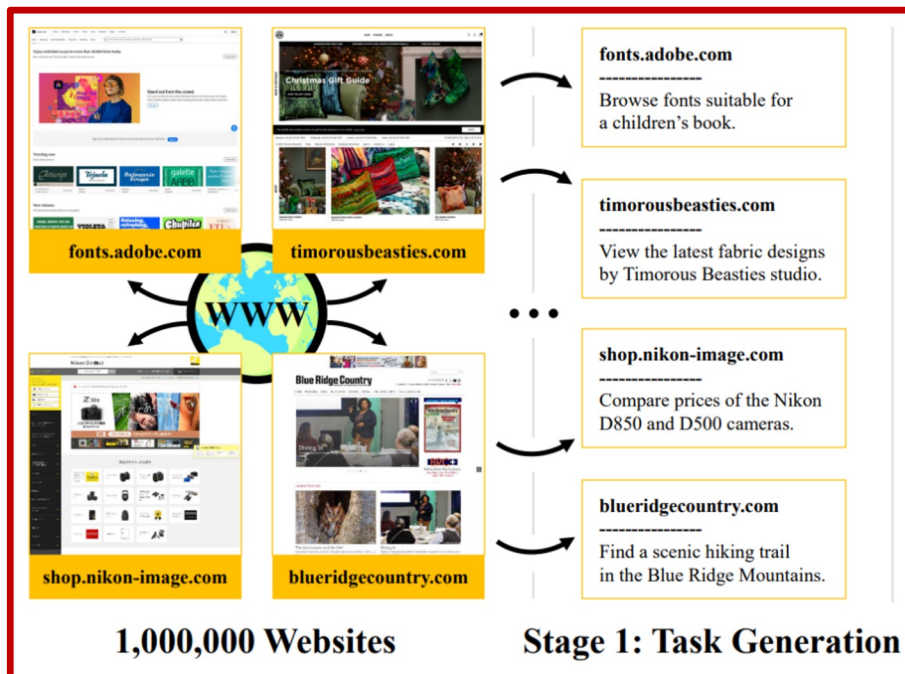
Scaling Task Generation

- Starting from **1M websites** we generate **150k challenging tasks**

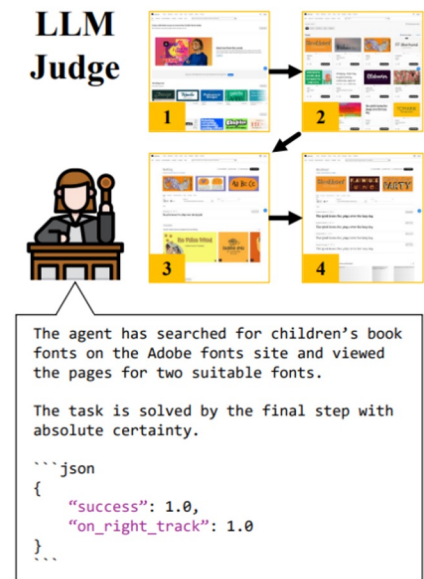


Revisiting The InSTA Pipeline

- **Key Idea:** LLMs can **generate and verify** agent tasks on live websites.



Stage 2: Attempt



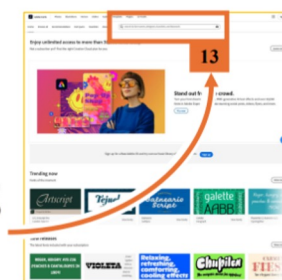
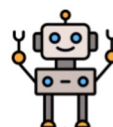
Stage 3: Evaluate

Revisiting The InSTA Pipeline

- **Key Idea:** LLMs can **generate and verify** agent tasks on live websites.
- We have **diverse and challenging** agent tasks.



LLM Agent

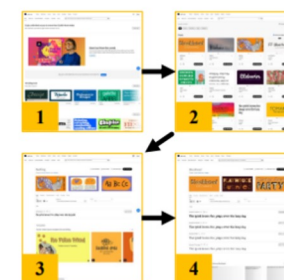


I'm viewing the Adobe fonts homepage, and I will type into the search box to browse fonts suitable for a children's book.

```
```json
{
 "action_key": "fill",
 "action_kwargs": {
 "text": "children's book"
 },
 "target_element_id": 13
}
```
```

Stage 2: Attempt

LLM Judge



The agent has searched for children's book fonts on the Adobe fonts site and viewed the pages for two suitable fonts.

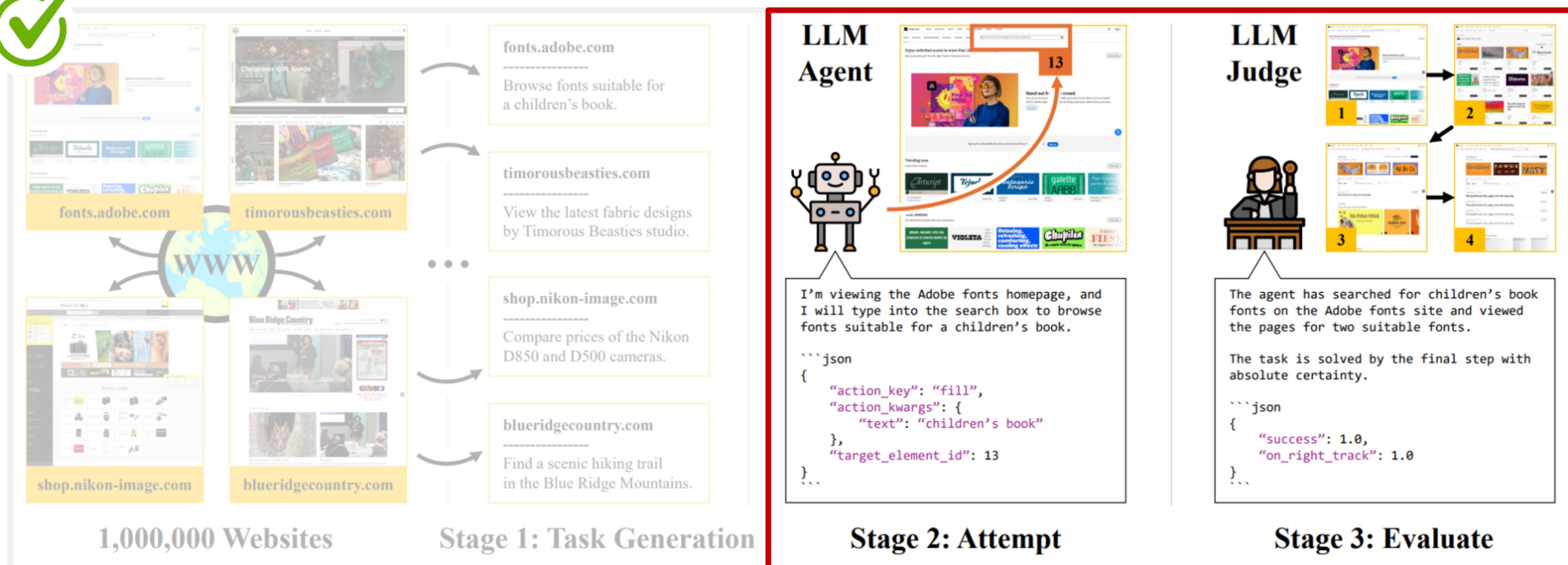
The task is solved by the final step with absolute certainty.

```
```json
{
 "success": 1.0,
 "on_right_track": 1.0
}
```
```

Stage 3: Evaluate

Revisiting The InSTA Pipeline

- **Key Idea:** LLMs can **generate and verify** agent tasks on live websites.
- Let's **evaluate current agents** on these tasks.



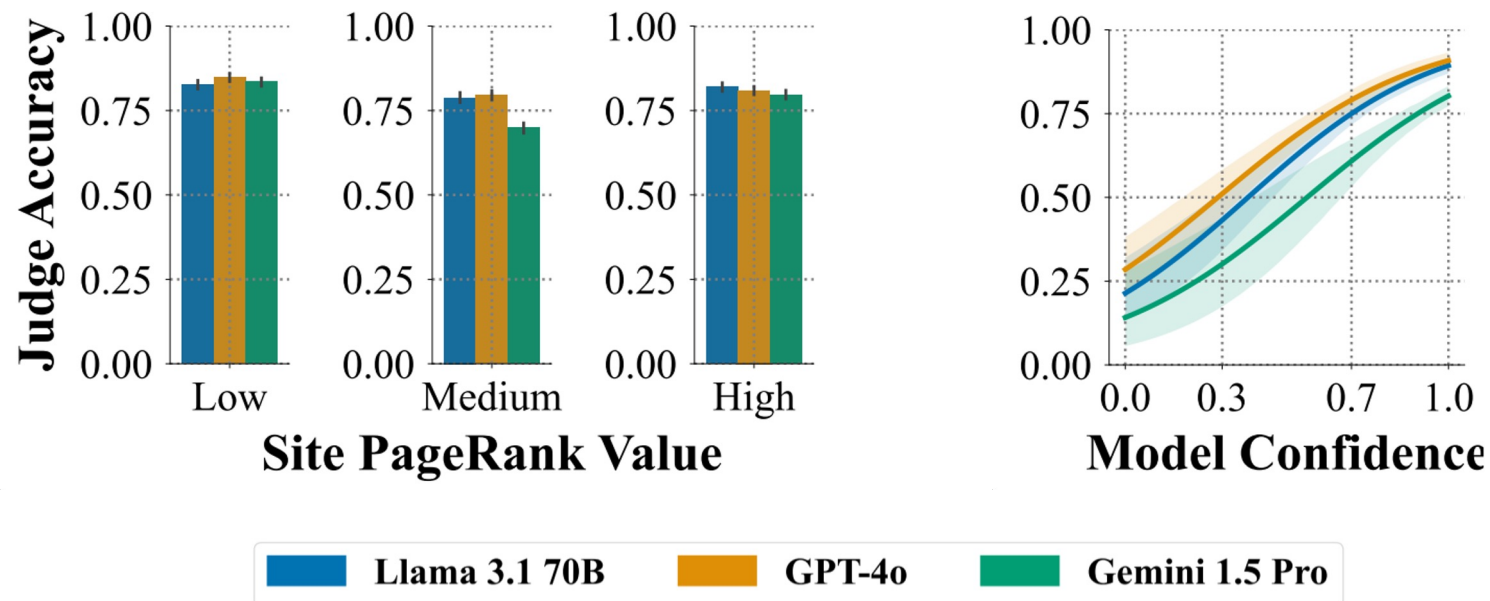
How Do We Evaluate Open-Ended Tasks?

- A simple strategy works for evaluation:
 - Observe task c , a sequence of actions, and the last observation
 - Estimate the **probability the task is a success** at the final step using LLM

$$\mathbf{r}_T = f^{\text{text} \rightarrow \text{val}} \left(\text{LLM}([\mathbf{y}_{\text{sys}}, \mathbf{c}, \mathbf{a}_1, \dots, \mathbf{a}_T, \text{Enc}(\mathbf{s}_T)]) \right)$$

- How robust are LLMs for **evaluating agent tasks**?

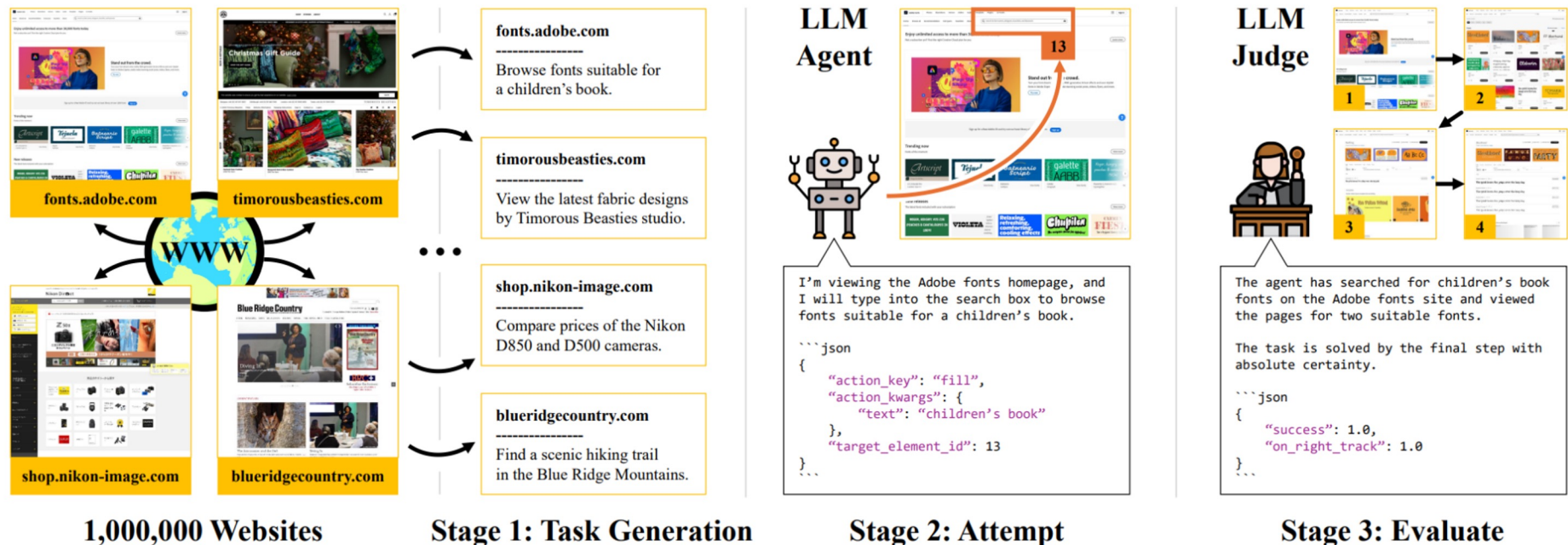
Language Models Are Robust Evaluators



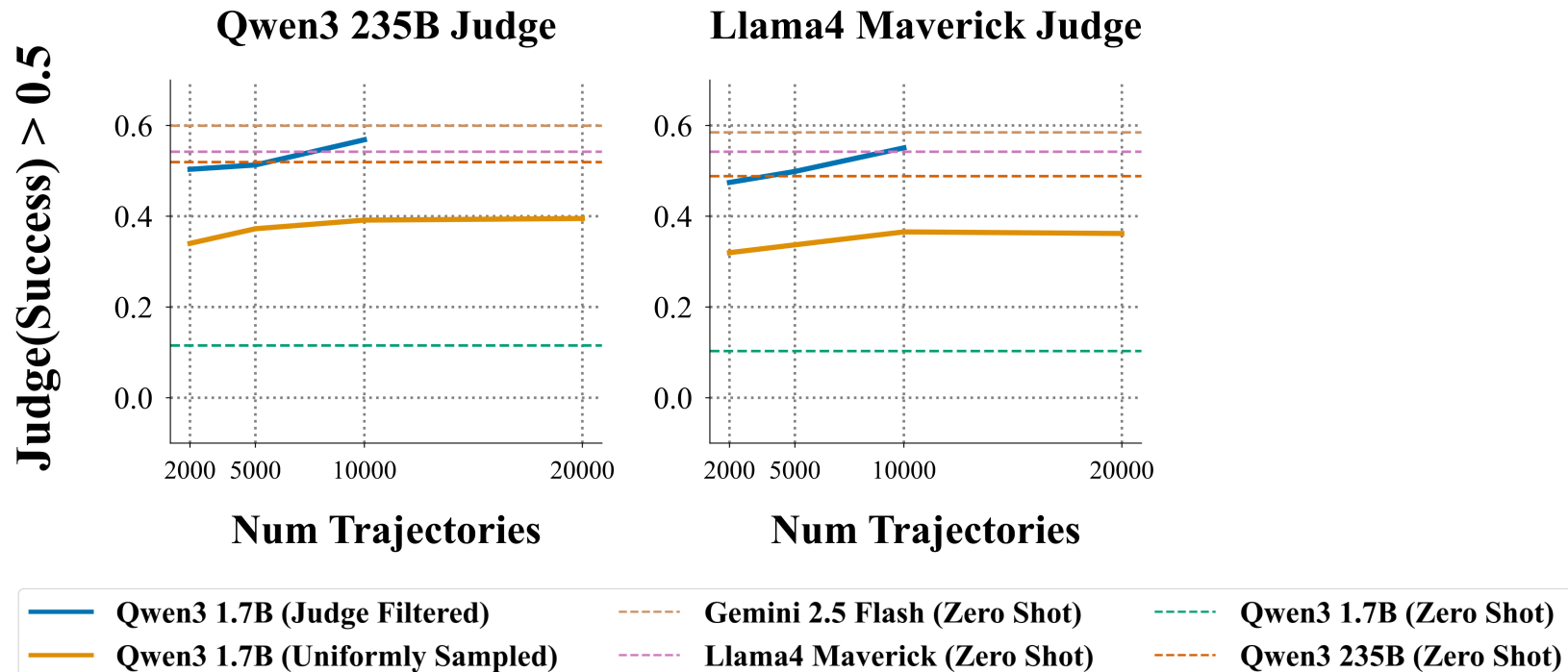
- LLMs **show high accuracy** for detecting successful trajectories.
- GPT-4o 82.6%, Llama 3.1 70B 81.7%, Gemini 1.5 Pro 78.0%

The InSTA Pipeline

- We can generate and verify agent tasks.
- What is the impact of scale for **training agents**?

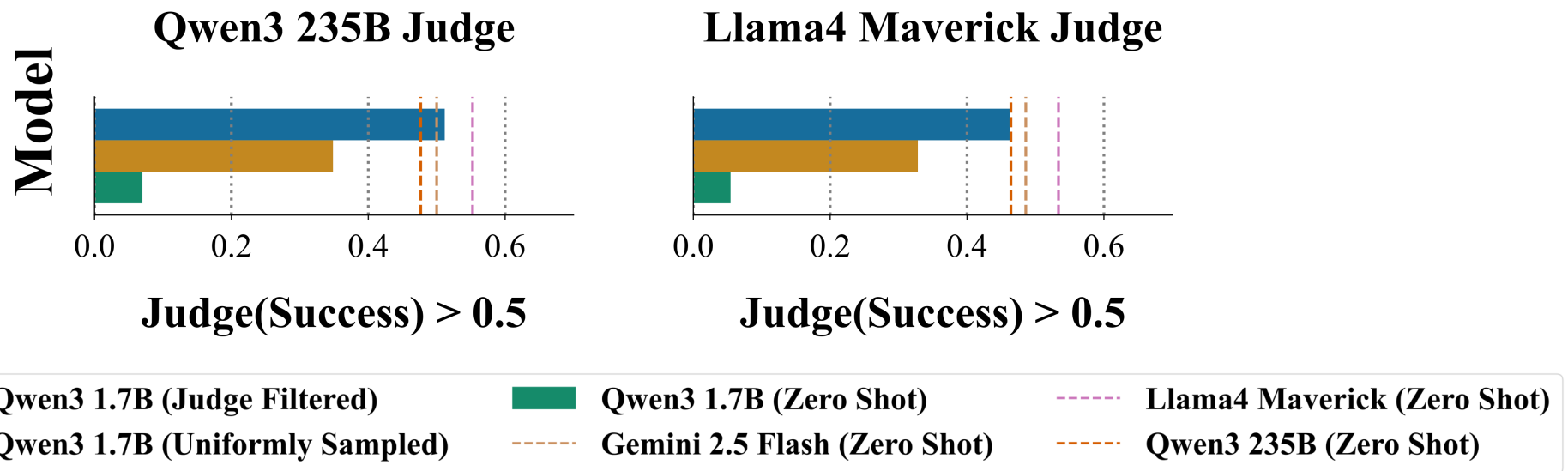


Scaling Training Data



- Qwen 3 1.7B outperforms Llama 4 Maverick by scaling data.
- +45.3% absolute percentage points on held-out websites.

Zero-shot transfer to WebVoyager



- Fine-tuned Qwen 3 1.7B matches frontier LLMs in performance

Next Steps

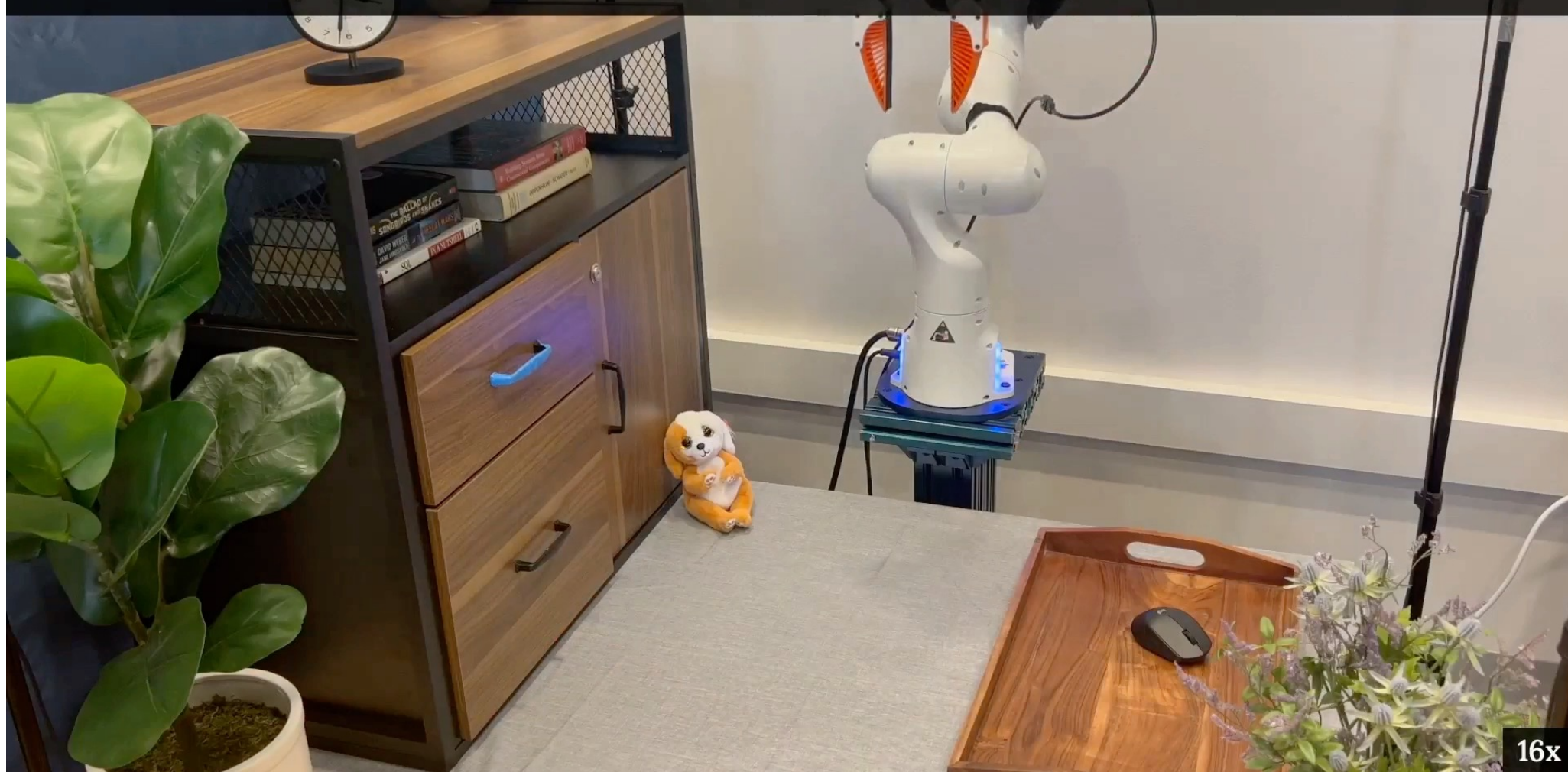
- There are 385M unique domains in the Common Crawl PageRank, suggesting another 1000x more data could be available by scaling further
- Moving towards **online RL**
- We plan to releasing a large dataset for multimodal agents from this study. Our dataset contains 2.2M screenshots, 2.2M CoT traces for actions, 150k traces for judge evaluations.
- From **Digital Agents** to **Physical Agents**

Long-Horizon Kitchen Task (10 stages): Autonomous Policy Rollout



Task: put the **mouse** in the drawer and close it

Environment: CabinetStore, Success Rate: 90%, #Stages: 4



16x

Manipulate novel objects with unseen receptacles



Summary

- VisualWebArena: a benchmark of realistic tasks designed to rigorously evaluate and advance the capabilities of autonomous multimodal web agents
- Inference-time search algorithm designed to enhance the capabilities of language model agents on realistic web tasks
- Data pipeline for large-scale generation and verification of synthetic web tasks, powered by Llama models
- **AI Safety and robustness, especially in the age of autonomous systems.**